

Pricing Exotic Options using Monte Carlo Techniques

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Abstract

We present a Bayesian calibration of exotic option pricing models using both traditional Markov Chain Monte Carlo and Hamiltonian Monte Carlo samplers in a unified framework. In our study, an Asian option is priced under two alternative dynamics, a Merton jump-diffusion model and a Heston stochastic volatility model, while a Margrabe exchange option is priced under a tqo-asset Black-Scholes model. Full Bayesian inference is performed for all model parameters, and model fit is evaluated via posterior predictive checks, Greek sensitivity analyses, residual error diagnostics, and price surface comparisons to a benchmark model. The HNC algorithm is applied to efficiently sample the posterior for the continuous parameter models, whereas adaptive Metropolis-within-Gibbs and birth-death Metropolis-within-Gibbs are applied for the non-HMC calibrations. The resulting posterior distributions reveal that the Heston model calibrated with HMC provides a better fit to the Asian option data over the Merton jump diffusion model, capturing some of the implied volatility skew and yielding better credible intervals for model prices compared to the Merton jump diffusion model. The Merton jump diffusion model exhibits a wide posterior uncertainty bands, reflecting greater uncertainty in parameter estimates. For the Margrabe exchange option, the simple two-asset Black-Scholes model achieves a near perfect fit under both non-HMC and HMC samplers, indicating little benefit from additional model complexity in that case. Model selection criteria (Bayesian and Deviance Informtation Criteria) further support these findings, with the Heston model being preferred over the Merton jump diffusion for the Asian option, while the Black-Scholes model is confirmed to be sufficient for the exchange option. These results show the effectiveness of Bayesian calibration for less complex models such as a multi-asset Black-Scholes in capturing essential market features and quantifying uncertainty in exotic option pricing models.

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1 Introduction

Quantitative finance and algorithmic trading has expanded dramatically over the last two decades. Automated, model-driven strategies now dominate many markets. For example, an estimated 70% of U.S. equity trading volume is initiated through algorithmic trading (Yadav, 2015). Global algorithmic trading activity continues to grow at double-digit rates annually, which is leading to a reliance on complex models. Options and other derivatives markets have similarly surged in activity. Cboe Global Markets reported a record 3.8 billion options contracts traded in 2024 (averaging 14.95 million contracts daily), marketing the fifth consecutive year of record-breaking volume.[1]

Quantitative hedge funds and trading firms that levrage these models have achieved remarkable success. Industry leaders like Renaissance Technologies and Marshall Wace delivered double-digit returns (over 20%) in 2024, outperforming many traditional asset managers. In fact, Marshall Wace's Eureka fund earned approximately 14.3%, and its Market Neutral Tops fund returned 22.6%, and its Alpha Plus fund gained 15.9% in 2024.[2] The scale of assets under management (AUM) at top quantitative firms has grown according, Citadel LLC manages about $397 billion as of August 2024,[3] and its market-making arm (Citadel Securities) reportedly generated a record $9.7 billion in trading revenue in 2024, a 55% increase from last year.[4] Such figures reflect not only impressive performance, but also the breadth of algorithmic option trading in today's markets. In particular, leading firms have built sophisticated option pricing engines (usually by stochastic models) that take advantage of small pricing inefficiencies at massive scale. This fusion of mathematical finance and computatiuon has been a key driver behind the success of quant hedge funds and the efficiency of derivative markets.

Despite these advances, traditional approaches to option pricing and model calibration face significant challenges when confronted with the complexity of real markets. Classical parameter estimation techniques such as *maximum likelihood estimation (MLE)* or least-squares calibration to market option prices encounter computational and statistical difficulties. Complex option pricing models that have stochastic volatility or jump processes, for example, have intractable likelihood functions that require integrating over unobserved state variables (like the entire path of volatility), which quickly becomes impractical in high dimensions (Brouillon et al., 2024). As the number of model parameters grows, conventional optimisation methods struggle. For example, the likelihood surfaces can be highly non-convex with multiple local optima, and it is difficult to establish convergence that guarantees for finding the true optimum. In practice, calibrating advanced models often demands brute-force searches or problem-specific approximations, which become computationally expensive as model complexity increases. Moreover, classical calibration typically creates point estimates that do not measure the uncertainty in the parameters. This is problematic in risk management and inference, where understanding the range of plausible parameter values and their joint confidence region is as important

[1] https://ir.cboe.com/news/news-details/2025/Cboe-Global-Markets-Reports-Trading-Volume-for-December-and-Full-Year-2024/

[2] https://www.reuters.com/markets/hedge-funds-deliver-double-digit-returns-2024-2025-01-02/#:~:text=British%20hedge%20fund%20Marshall%20Wace,Eureka%20fund%2C%20a%20source%20said.

[3] https://www.investopedia.com/articles/personal-finance/011515/worlds-top-10-hedge-fund-firms.asp

[4] https://www.bloomberg.com/news/articles/2025-03-06/citadel-securities-9-7-billion-trading-revenue-passes-barclays

as finding a single best fit.

Similar issues arise on the pricing side with classical deterministic techniques. *Partial differential equation (PDE) solves* and lattice methods such as *binomial/trinomial tress* provide elegant solutions for simple option models, but they suffer from *the curse of dimensionality*. The dimensionality of a PDE grows with the number of underlying risk factors of path dependencies, and standard grid-based methods become infeasible beyond a few dimensions (Han et al., 2018). This curse of dimensionality exponentially grows in computational cost with each additional underlying, making classical PDE approaches impractical for multi-asset or path-dependent options (such as American options). Although closed-form pricing formulas exist for a few idealised cases (such as *Black-Scholes* for European options), many exotic options have no analytic solution. Traders often resort to *Monte Carlo simulation* for such cases, however, straightforward Monte Carlo can converge slowly and embedding it inside an optimisation loop for calibration increases the computation load. Additionally, likelihood based inference for stochastic volatility models may require integrating out latent variables at each step, which closed-form solutions are rarely available for. These limitations of classical methods show that there is a need for more flexible, simulation-based approaches that can handle high-dimensional integrations and complex posterior spaces.

Over the past two decades, *Bayesian methods* powered by *Markov Chain Monte Carlo (MCMC)* have increased as an alternative for inference in quantitative finance. In a Bayesian framework, we treat model parameters and latent states as random variables with a prior distribution, and update this to a posterior distribution given observed data (such as historical prices). MCMC algorithms are then used to sample from these posterior distributions, which effectively solves inference problems by simulation rather than by analytical integration. MCMC evaluates the model's likelihood function (up to a constant) and the prior, it does not require closed-form solutions for the posterior or simplifying linear approximations. This approach avoids many of the problems of classical methods such as high-dimensional integrations, which is handled by random sampling, and there is no need to linearise nonlinear models to derive explicit formulae for complex likelihoods. In the context of financial models, researchers have found that Bayesian and MCMC methods are popular in option pricing models, especially those extending the Black-Scholes-Merton framework to include jumps and stochastic volatility.

The fundamental MCMC technique is the *Metropolis-Hsting (MH)* algorithm, which is often combined with *Gibbs sampling* for models that include latent variables. Each MH step proposes a change in some parameter or state and accepts or rejects it based on the *Metropolis-Hastings acceptance probability* to ensure the chain converges to the true posterior. Through these iterative simulations, the MCMC approach builds up a representative sample from the posterior distribution of all unknown quantities. The outcome is an entire distribution of parameter values consistent with the data, from which credibility intervals and other measures of uncertainty can be derived. This ability to measure parameter uncertainty is a major advantage of the Bayesian MCMC approach over classical point estimation.

Over time, researchers have developed a variety of enhanced MCMC techniques to improve efficiency and convergence in financial applications. One important class is *adaptive MCMC* algoirthms, which automatically tune their proposal distributions per iteration (Haario et al., 2001). Such adaptive schemes help in high-dimensional or strongly corre-

lated parameter spaces, where a fixed proposal might either mix too slowly or be prone to rejection. Another powerful innovation is *Hamiltonian Monte Carlo (HMC)*, which leverages gradient informaiton from the target density to guide the sampling. HMC introduces auxiliary momentum variables and uses *Hamiltonian dynamics* to propose distant, yet high-probability, moves in the parameter space, which dramatically reduces the random walk behaviour of traditional MCMC (Hoffman et al., 2014). HMC can produce more effective samples per iteration by weakening the dependency between successive samples and imporving exploration efficiency. This feature is especially useful in complex posterior spaces such as those in high-dimensional stochastic volatility models. These samplers, along with other MCMC techniques like *particle MCMC* (which combines particle filters with MCMC for state-space models) (Andrieu et al., 2010) and *reversible jump MCMC*, have greatly expanded the toolkit for Bayesian computation in finance. Applications have ranged from Bayesian estimation of Heston's stochastic volatility model and Merton's jump diffusion model to filtering problems in interest rate and credit models.

In this paper, we build on these developments to study MCMC methods in exotic option pricing. We present a self-contained formulation and empirical evaluation covering both well-known option pricing models and modern MCMC techniques, applied to equity and commodity options. We begin with laying out the theoretical framework for a range of option pricing models. starting with the classic Black-Scholes model and then consider progressively richer models, including Merton's jump-diffusion model and Heston's stochastic volatility model. We additionally include the foundational knowledge of *Birth Death processes*, which will play an important role later in the paper. We then introduce the Bayesian inference approach and the specific MCMC algorithms employed. We formulate a Bayesian estimation problem for the parameters of the aforementioned models, given observed data. We specify prior distributions for model parameters and derive the likelihood functions for each model. The construction of the MCMC sampler is then described, including the choice of proposal kernels and any augmentation with latent variables. In particular, we discuss how latent volatility paths or latent jump indicators are introduced to facilitate inference in the Heston and jump-diffusion models, respectively. We employ a *Metropolis-Hastings within Gibbs sampler* for most parameters along with *adaptive proposal tuning* to achieve reasonable acceptance rates. For the Heston model, we also experiment with a Hamiltonian Monte Carlo step to efficiently sample the highly correlated volatility parameters. The goal of this section is to provide a clear blueprint of the inferential procedure, which we will integrate with the option pricing models of Section 2. Section 4 conducts analytical valuation of the exotic contracts considered in our study. We collect closed-form and transform-based results for geometric Asian options and Margrabe's exchange option under the models of Section 2. In particular, we derive the Black-Scholes lognormal formulas, the Merton mixture/conditional-normal representation, and the Heston formulas obtained via *Feynman-Kac arguments* (for Asians) (Shreve et al., 2004) and Fourier mthods based on the characteristic function of the integrated variance (for Margrabe) (Heston, 1993). Finally, we conclude this paper with tests on the MCMC algorithms on empirical data. First, we perform controlled experiments with synthetic exotics (Asian and Margrabe) to verify identifiably and coverage. Option values are generated from one model while inference is carried out under an alternative, and we examione posterior recovery, credible sets, and posterior predictive pricing errors, using corn (CORN) and soya bean (SOYB) data.

2 Markov Chain Monte Carlo Techniques

2.1 Metropolis Hastings

Take a Black-Scholes market with one risky asset, modeled as:

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t,$$

where $\{W_t\}_{t\geq 0}$ is the Wiener process. By Girsanov's theorem, under the risk-neutral measure $\mathbb{Q}$, we model the stock as

$$\frac{dS_t}{S_t} = r dt + \sigma dW_t.$$

By *Ito-Lemma*, we can obtain the log prices X_t:

$$dX_t = \left(r - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t.$$

We here apply *Euler's discretisation* to

$$X_{t+\Delta t} = X_t + (r - \frac{\sigma^2}{2})\Delta t + \sigma\sqrt{\Delta t} Z_t,$$

$$X_{t+\Delta t}|X_t \sim \mathcal{N}\left(\left(r - \frac{\sigma^2}{2}\right)\Delta t, \sigma^2\Delta t\right).$$

Here we can take the Loglikelihood with respect to the parameters $\boldsymbol{\theta}$ as:

$$\begin{aligned}\mathcal{L}(\boldsymbol{X}|\boldsymbol{\theta}) = \log \ell(\boldsymbol{X}|\boldsymbol{\theta}) &= \sum_{t=0}^{T-1} \log \mathbb{P}(X_t|\boldsymbol{\theta}) \\ &= T \log\left(\frac{1}{\sqrt{2\pi\sigma^2\Delta t}}\right) - \sum_{t=0}^{T-1} \frac{\left(X_{t+\Delta t} - X_t - \left(r - \frac{\sigma^2}{2}\right)\Delta t\right)^2}{2\sigma^2\Delta t}.\end{aligned}$$

Since volatility is non-negative ($\sigma > 0$), we take the log prior as the log normal:

$$\begin{aligned}\sigma \sim \text{LogNormal}(m, s^2), \quad \log \pi(\sigma) &= \log\left[\frac{1}{\sigma\sqrt{2\pi s^2}} \exp\left[-\frac{(\ln\sigma - m)^2}{2s^2}\right]\right] \\ &= -\ln\sigma - \frac{1}{2}\ln(2\pi s^2) - \frac{(\ln\sigma^2 - m)^2}{2s^2}.\end{aligned}$$

Hence, by taking $\boldsymbol{\theta} = \sigma$, the log posterior, with $c \in \mathbb{R}$ constant, is:

$$\log \pi(\sigma|\boldsymbol{X}) = \mathcal{L}(\boldsymbol{X}|\sigma) + \log \pi(\sigma) + c.$$

Now suppose that we have K iterations in our MCMC algorithm and $\sigma^{(k)}$ at the k iteration with propose σ' from a proposal density $q(\sigma'|\sigma^{(k)})$. Then the Metropolis Hastings acceptance probability is

$$\alpha(\sigma^{(k)}, \sigma') = \min\left(1, \frac{\pi(\sigma'|\boldsymbol{X})q(\sigma^{(k)}|\sigma')}{\pi(\sigma^{(k)}|\boldsymbol{X})q(\sigma'|\sigma^{(k)})}\right),$$

which can be written in log form as

$$\log \alpha(\sigma^{(k)}, \sigma') = \min\left(0, \log \pi(\sigma'|\boldsymbol{X}) + \log q(\sigma^{(k)}|\sigma') - \log \pi(\sigma^{(k)}|\boldsymbol{X}) - \log q(\sigma'|\sigma^{(k)})\right).$$

The acceptance probability allows us to calculate the *empirical acceptance rate* for the algorithm. This allows us to determine how many proposed $\boldsymbol{\theta}$ terms are accepted through all iterations. At iteration k, we propose (for the Black-Scholes model), $\sigma' \sim q(\cdot|\sigma^{(k)})$ and accept it with probability $\alpha(\sigma^{(k)}, \sigma')$. We then draw a uniform $u \sim \text{Uniform}(0, 1)$, such that if $u < \alpha(\sigma^{(k)}, \sigma')$ we accept the proposal and make $\sigma^{(k+1)} = \sigma'$. Otherwise, we reject the proposal and set $\sigma^{(k+1)} = \sigma^{(k)}$. This then leads us to the empirical acceptance rate:

$$AR = \frac{1}{K} \sum_{k=1}^{K} \mathbf{1}\{\text{move at } k \text{ was accepted}\}.$$

It is important to note that in practical cases, we aim to have the acceptance rate around 0.2-0.4.

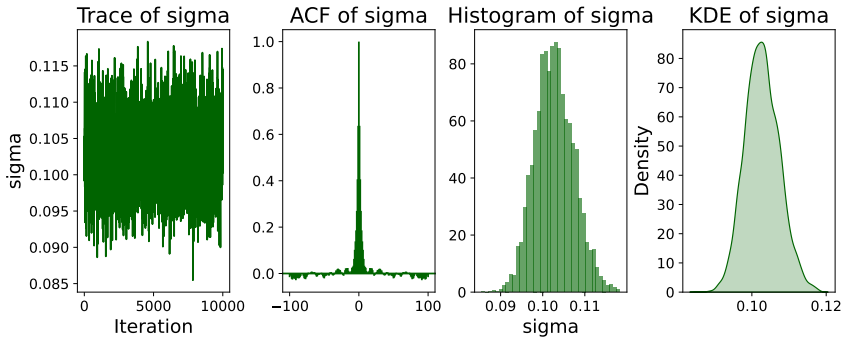


Figure 1: Diagnostic plots for the random walk Metropolis sampler for σ: (i) trace plots, (ii) autocorrelation functions (ACF), (iii) histograms, and (iv) kernel density estimates (KDE). The trace shows stationary exploration with the ACF decaying rapidly to near zero. The distribution are unimodal and approximately Gaussian.

2.1.1 Effective Sample Size (ESS)

When working with MCMC algorithms, the successive MCMC draws $\{\boldsymbol{\theta}^{(k)}\}_{k=1}^{K}$ are serially correlated. The *Effective Sample Size (ESS)* allows us to quantify how many independent samples the chain is worth. Recall that for a sequence of independent identically distributed samples from π, $\boldsymbol{\theta^{(1)}}, \ldots, \boldsymbol{\theta}^{(K)}$, if we set $h(\boldsymbol{\theta})$ as a scalar summary, then we can define the *lag-t autocorrelation* as

$$\rho_t = \frac{\mathbb{C}\text{ov}(h(\boldsymbol{\theta}^{(i)}), h(\boldsymbol{\theta}^{(i+t)}))}{\mathbb{V}\text{ar}(h(\boldsymbol{\theta}))}.$$

We then define the ESS for $h(\boldsymbol{\theta})$ as

$$ESS = \frac{K}{1 + 2\sum_{t=1}^{L} \rho_t}.$$

Typically we examine the ESS for each component of $\boldsymbol{\theta}$ to ascertain whether sufficient mixing has occurred. A good ESS is one that allows us to calculate what you need to the

Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	0.1027	0.0045	2 134.5112

Table 1: Posterior summary for the Black-Scholes volatility parameter under a random walk Metropolis sampler. The table shows posterior mean, posterior standard deviation, and univariate ESS for σ. The posterior is concentrated around $\hat{\sigma} \approx 0.103$ with small dispersion, and the ESS indicates moderate dependence typical of diffusion type proposals on a well identified one dimensional target.

required accuracy. Often an ESS of 500 is deemed sufficient, although an ESS below 100 means that inference from the MCMC sample should be treated with caution.

Figure 1 shows the geometry of the Black-Scholes likelihood and the local nature of a random walk proposal. Under discretely observed log-returns, the log likelihood for σ is close to quadratic in a neighbourhood of its maximiser, so the posterior for σ is well approximated by a normal law on the working scale. This is visible in the distribution plots which are unimodal and nearly symmetric with small variance, and in the trace, which remains confined to a narrow band about 0.103 while exhibiting frequent level crossings after burn-in. The ACF shows a positive short-lag correlation that dissipates over moderate lags, which yields an ESS of roughly 2.1×10^3 in table 1. This displays a well-tuned one-dimensional random walk that nonetheless explores by local diffusion rather than by long, directed moves.

2.2 Metropolis-within-Gibbs

When applying MCMC to models with many parameters and latent variables, we apply *Gibbs sampling*. Gibbs sampling iteratively samples each parameter from its conditional distribution given all other variables. As such, we iteratively update latent states in blocks based on these distributions. However, when we do not have a closed form distribution, we apply Metropolis-Hastings for that block. This is known as *Metropolis-within-Gibbs*. The idea is we cycle through components of $\boldsymbol{\theta} = (\theta_1, \theta_2, \ldots, \theta_p)$, and for each θ_i we perform a MH update to get the conditional density $\pi(\theta_i \mid \theta_{-i})$, for $i = 1, ..., p$, which will given the full joint posterior $\pi(\theta_1, ..., \theta_p)$ invariant. We use Gibbs for the rest of our pricing models due to the high dimensionality of both the options and complexity of the models.

Choosing a block is flexible, we can update one parameter at a time, or a group of highly correlated parameters together. For example, in the Heston model, we can use a Gibbs strategy that alternatives between sampling the volatility latent path $v_{0:T}$ given the parameters, and sampling the parameters $(\kappa, \theta, \xi, \rho)$ given the volatility path and price data. If we take the Heston model (without volatility jumps) below:

$$\begin{aligned} dS_t &= \mu S_t dt + S_t \sqrt{v_t} dW^{\mathbb{P}}_{t,2}, \\ dv_t &= \kappa(\xi - v_t)dt + \sigma\sqrt{v_t} dW^{\mathbb{P}}_{t,2}, \\ d\langle W^{\mathbb{P}}_{t,1}, W^{\mathbb{P}}_{t,2}\rangle &= \rho dt, \end{aligned}$$

where $2\kappa\xi > \sigma^2$, $\kappa, \sigma, \xi, \sqrt{v_t} > 0$, and $\rho \in (-1, 1)$. We can take $\boldsymbol{\theta} = (\kappa, \sigma, \xi, \rho)$By applying

Girsanov's theorem[5], we convert into risk neutral measure:

$$\begin{aligned} dS_t &= rS_t dt + \sqrt{v_t} S_t dW_t^{(1)}, \\ dv_t &= \kappa(\xi - v_t)dt + \sigma\sqrt{v_t} dW_t^{(2)}, \\ d\langle W_t^{(1)}, W_t^{(2)} \rangle &= \rho dt, \end{aligned}$$

where $r > 0$ is the risk-free rate, and $W_t^{(1)}, W_t^{(2)}$ are the Wiener processes under the risk neutral measure $\mathbb{Q}$. Here the logarithmic prices $X_t = \ln S_t$ are:

$$dX_t = \left(r - \frac{v_t}{2}\right) dt + \sqrt{v_t} dW_t^{(1)}$$

Under the Euler discretisation, the discrete time approximation increments are:

$$\Delta X_t := X_{t+1} - X_t, \quad \Delta v_t := v_{t+1} - v_t,$$

are approximately jointly normal with mean

$$\mathbb{E}[\Delta X_t] = \left(r - \frac{1}{2} v_t\right) \Delta t = \mu_{X,t} \Delta t, \quad \mathbb{E}[\Delta v_t] = \kappa(\xi - v_t)\Delta t = \mu_{v,t} \Delta t,$$

with the covariance matrix

$$\Sigma_t = \begin{pmatrix} v_t \Delta t & \rho\sigma v_t \Delta t \\ \rho\sigma v_t \Delta t & \sigma^2 v_t \Delta t \end{pmatrix} = \begin{pmatrix} \mathbb{V}\mathrm{ar}(X_t) & \mathbb{C}\mathrm{ov}(X_t, v_t) \\ \mathbb{C}\mathrm{ov}(X_t, v_t) & \mathbb{V}\mathrm{ar}(v_t) \end{pmatrix},$$

hence the joint PDF:

$$f(\Delta X_t, \Delta v_t, \boldsymbol{\theta}) = \frac{1}{2\pi\sqrt{\det \Sigma_t}} \exp\left(-\frac{1}{2} \left(\begin{pmatrix} \Delta X_t \\ \Delta v_t \end{pmatrix} - \begin{pmatrix} \mu_{X,t} \\ \mu_{v,t} \end{pmatrix} \right)^T \Sigma_t^{-1} \left(\begin{pmatrix} \Delta X_t \\ \Delta v_t \end{pmatrix} - \begin{pmatrix} \mu_{X,t} \\ \mu_{v,t} \end{pmatrix} \right) \right) \tag{1}$$

We can rewrite the joint density as:

$$f(\Delta X_t, \Delta v_t) = f(\Delta X_t | \Delta v_t) f(\Delta v_t),$$

and if we expand (1), we can rewrite it such that:

$$f($$

Hence,

$$\begin{aligned} \Delta X_t \mid \Delta v_t &\sim \mathcal{N}\left(\mu_X + \frac{\mathbb{C}\mathrm{ov}(X_t, v_t)}{\mathbb{V}\mathrm{ar}(X_t)}, \quad \mathbb{V}\mathrm{ar}(v_t) - \frac{(\mathbb{C}\mathrm{ov}(X_t, v_t))^2}{\mathbb{V}\mathrm{ar}(X_t)} \right), \\ \Delta v_t | \Delta X_t &\sim \mathcal{N}\left(\mu_v + \frac{\mathbb{C}\mathrm{ov}(X_t, v_t)}{\mathbb{V}\mathrm{ar}(v_t)}, \quad \mathbb{V}\mathrm{ar}(X_t) - \frac{(\mathbb{C}\mathrm{ov}(X_t, v_t)^2}{\mathbb{V}\mathrm{ar}(v_t)} \right), \end{aligned} \tag{2}$$

Thus our likelihood for both X and v respectively is:

When calculating the priors for $\boldsymbol{\theta} = (\sigma, \kappa, \xi, \rho)$ we have to take into account the constraints of the parameters. Each parameter in the Heston model follow the following constraints:

$$\sigma > 0, \quad \kappa > 0, \quad \xi > 0, \quad \rho \in (-1, 1)$$

[5]See appendix A

Hence, we have to adopt the bijections:

$$\begin{aligned}\sigma &= \exp(\theta_1),\\ \kappa &= \exp(\theta_2),\\ \xi &= \exp(\theta_3),\\ \rho &= \tanh(\theta_4),\end{aligned}$$

where each $\theta_i \in \mathbb{R}$ is unconstrained. Hence our inverse mapping is

$$\theta_1 = \ln(\sigma), \quad \theta_2 = \ln(\kappa), \quad \theta_3 = \ln(\xi), \quad \theta_4 = \frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right).$$

If we want to transform the densities from $\boldsymbol{\theta}_{unc}$ back to $\boldsymbol{\theta}$, then we require the Jacobian determinant:

$$\left|\det\left(\frac{d\boldsymbol{\theta}}{d\boldsymbol{\theta}_{unc}}\right)\right| = \left|\det\begin{pmatrix} \frac{\partial\sigma}{\partial\theta_1} & 0 & 0 & 0 \\ 0 & \frac{\partial\kappa}{\partial\theta_2} & 0 & 0 \\ 0 & 0 & \frac{\partial\xi}{\partial\theta_3} & 0 \\ 0 & 0 & 0 & \frac{\partial\rho}{\partial\theta_4} \end{pmatrix}\right|.$$

This gives us the following

$$\left|\det\left(\frac{d\boldsymbol{\theta}}{d\boldsymbol{\theta}_{unc}}\right)\right| = \sigma\kappa\xi(1-\rho^2).$$

Interestingly, the Jacobian matrix cancels out if we place our priors on $\boldsymbol{\theta}_{unc}$ directly, and hence doesn't appear in the acceptance ratio. This two-block Gibbs can be broken down further by breaking down the volatility path into smaller segments, such that we update (in each iteration) each individual volatility path given the parameters and data. This is known *Single-site Metropolis*.

Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	0.0024	0.0252	111.6038
κ	8 297 784.4025	9 960 223.6531	2.8301
ξ	7 909 900 299 532.2783	11 693 122 264 912.1875	3.2745
ρ	-0.8547	0.3487	7.8790

Table 2: Posterior summary for the Heston parameter vector $\boldsymbol{\theta} = (\sigma, \kappa, \xi, \rho)$ under the Metropolis-within-Gibbs sampler with single-site updates of the latent volatility path. The table shows the posterior means, posterior standard deviations, and univariate ESS. The low ESS for κ, ξ, and ρ indicates substantial dependence and poor mixing in those spaces.

In figure 2, we can see that the posterior geometry is highly anisotropic across all coordinates. The σ trace remains stable on a moderate positive scale and its histogram and KDE are right-skewed, which is a direct consequence of the positive mapping $\sigma = \exp(\theta_1)$. It's ACF decays faster than for the other variables, implying comparatively better mixing for this diffusion-scale parameter. In contrast, we can see that κ is more tightly concentrated on [0, 0.4] and the ACF exhibits visible persistence, which together with the relatively

flat KDE over this narrow range, displays a strong local correlation and slow local moves in κ. Additionally, ξ shows exploration across magnitudes of 0 to approximately 450, with a distinctly heavy-tailed distribution and a slowly decaying ACF, indicating that the random walk behaviour is on a poorly identified scale direction. ρ is bounded between (-1, 1) but spends considerable time on the negative side. Since $\rho = \tanh(\theta_4)$, the metric compresses near ± 1, so equal sized proposals in θ_4 translate into vanishing moves in ρ as $\rho \to 1$, producing boundary stickiness that is visible both in the trace and the ACF.

Although placing priors on the unconstrained θ cancels the Jacobian in the acceptance ratio, the mapping still induces right-skew and scale heterogeneity in (σ, κ, ξ), and strong posterior coupling between κ and ξ, which single-site state updates propagate slowly through the latent path. The result is high autocorrelation and low effective sample sizes for (κ, ξ, ρ).

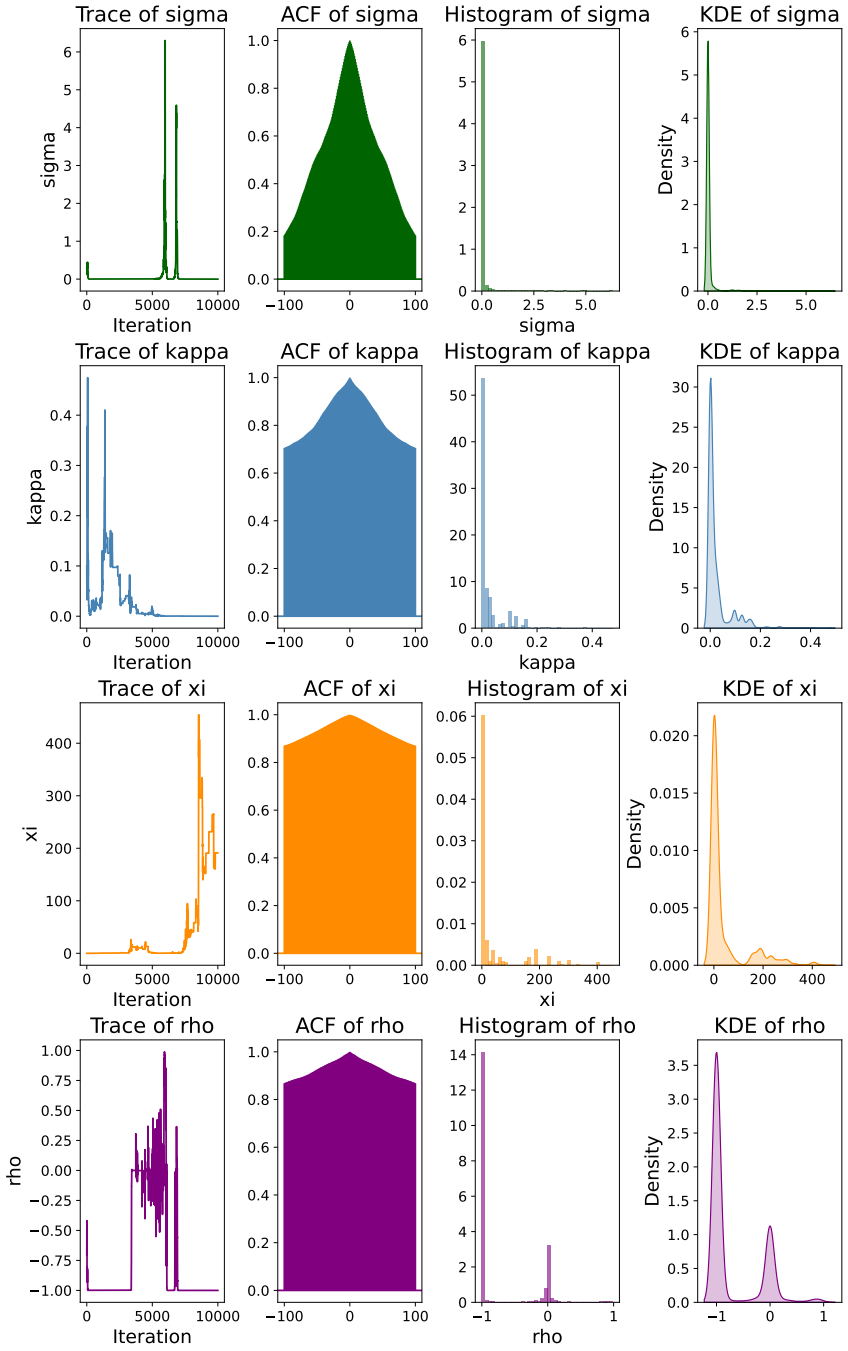


Figure 2: Diagnostic plots for each Heston parameter under the Metropolis-within-Gibbs sample: (i) trace plots, (ii) autocorrelation functions (ACF), (iii) histograms, and (iv) kernel density estimates (KDE). The σ plots show a strictly positive support with a right-skewed marginal, whereas κ is numerically compressed to the interval [0, 0.4] with a persistent autocorrelation. ξ explores a scale approximately up to 450 with a heavy right tail and ρ evolves within (-1, 1) with mass concentration on the negative side. The traces and slowly decaying ACFs indicate sluggish exploration for (κ, ξ, ρ) relative to σ.

2.3 Adaptive Metropolis Hastings

A standard random walk metropolis algorithm uses a fixed proposal covariance, C. If C is too small, the chain mixes slowly (high autocorrelation). On the other hand, if too

large, proposals are often rejected. In high or moderate dimensions, optimal choices of C depend on the correlation structure of the target posterior.

We want to let the chain learn an approximately optimal proposal shape on each iteration by estimating the posterior's local covariance structure from past samples. The ideal proposal for a Gaussian target $\mathcal{N}(\mu, \Sigma)$ is $\mathcal{N}(\theta, \Sigma)$. *Adaptive-Metropolis Hastings AMH* seeks to mimic this by continually updating an empirical covariance, allowing the proposal distribution to evolve in response to information gathered from past samples, thereby automatically tuning itself to the shape of the target posterior.

Say we had a two risky assets S_1, S_2 under the Black-Scholes, with correlation $\rho \in (-1, 1)$, such that:

$$\begin{aligned} dS_{1,t} &= rS_{1,t}dt + \sigma S_{1,t}dW^{\mathbb{Q}}_{1,t}, \\ dS_{2,t} &= rS_{2,t}dt + \nu S_{2,t}dW^{\mathbb{Q}}_{2,t}, \\ d\langle W^{\mathbb{Q}}_{1,t}, W^{\mathbb{Q}}_{2,t}\rangle &= \rho dt, \end{aligned}$$

where $\sigma, \nu > 0$, and $W^{\mathbb{Q}}_{i,t}$ are the Wiener processes in the risk neutral measure for each risky asset $i = 1, 2$. If we let $\boldsymbol{\theta} \in \mathbb{R}^3$ denote the current state of the Markov chain, and let $\pi(\boldsymbol{\theta}|\boldsymbol{X})$ be the three dimensional posterior over (σ, ν, ρ) for the two-asset Black-Scholes model, reparameterised via $\theta_1 = \ln \sigma, \theta_2 = \ln \nu, \theta_3 = \operatorname{arctanh}(\rho)$.

At iteration n, given the current state $\boldsymbol{\theta}^{(n)}$ and a symmetric positive-definite proposal covariance $C^{(n)}$, the AMH algorithm generates a candidate $\boldsymbol{\theta}'$ by drawing from the Gaussian proposal distribution $\mathcal{N}(\boldsymbol{\theta}^{(n)}, C^{(n)})$. The acceptance probability for the Metropolis algorithm is then computed as

$$\alpha(\boldsymbol{\theta}^{(n)}, \boldsymbol{\theta}') = \min\left(1, \frac{\pi(\boldsymbol{\theta}'|\boldsymbol{X})q(\boldsymbol{\theta}^{(n)}|\boldsymbol{\theta}')}{\pi(\boldsymbol{\theta}^{(n)}|\boldsymbol{X})q(\boldsymbol{\theta}'|\boldsymbol{\theta}^{(n)})}\right) = \min\left(1, \frac{\pi(\boldsymbol{\theta}'|\boldsymbol{X})}{\pi(\boldsymbol{\theta}^{(n)}|\boldsymbol{X})}\right),$$

and $\boldsymbol{\theta}^{(n+1)}$ is set equal to $\boldsymbol{\theta}'$ with probability α, or remains at $\boldsymbol{\theta}^{(n)}$ otherwise. Upon accepting or rejecting this proposal, we update the empirical mean $M^{(n)}$ and sum of squares matrix $S^{(n)}$ using the *one-pass algorithms* (Welford, 1962):

$$\begin{aligned} M^{(n+1)} &= M^{(n)} + \frac{\boldsymbol{\theta}^{(n+1)} - M^{(n)}}{n+1} \\ S^{(n+1)} &= S^{(n)} + (\boldsymbol{\theta}^{(n+1)} - M^{(n)})(\boldsymbol{\theta}^{(n+1)} - M^{(n+1)})^\top, \end{aligned}$$

so that the empirical covariance is $\hat{\Sigma}^{(n+1)} = \frac{S^{(n+1)}}{n}\ \forall n \geq 1$. The idea of adaptation consists in blending the previous proposal covariance $C^{(n)}$ with a scaled version of the empirical covariance. Specifically, after a predetermined burn-in period of n_0 iterations, we define the scaled empirical covariance as

$$\Sigma^{(n+1)}_{\text{emp}} = \frac{(2.38)^2}{d}\hat{\Sigma}^{(n+1)} + \epsilon I_d,$$

where the factor $\frac{(2.38)^2}{d}$ is known from the optimal-scaling theory of Gaussian targets to make an asymptotic acceptance rate of approximately 23.8%, and $\epsilon > 0$ is a small ridge term to ensure eigenvalues remain bounded away from zero (Gelman et al., 1996). Hence, the new proposal covariance is then set to

$$C^{(n+1)} = \gamma C^{(n)} + (1 - \gamma)\Sigma^{(n+1)}_{\text{emp}},$$

with $\gamma \in (0, 1)$ chosen so that changes to the proposal kernel become smaller as n grows. For $n < n_0$, we let $C^{(n+1)} = C^{(n)}$. The two theoretical properties that guarantee the AMH chain remains ergodic with the correct stationary distribution are the *diminishing adaptation condition* and the *containment condition*. The former condition requires that the total-variation distance between successive proposal kernels tends to zero. In other words, if $K_n(\boldsymbol{\theta}, \cdot)$ is the proposal kernel at step n, then we require

$$\sup_{\boldsymbol{\theta}} ||K_{n+1}(\boldsymbol{\theta}, \cdot) - K_n(\boldsymbol{\theta}, \cdot)||_{\text{TV}} \longrightarrow 0 \quad \text{as } n \to 0.$$

This is ensured by the convex combination with weight γ and by the fact that empirical covariances converge (Haario et al., 2001). The containment condition requires that the family of proposal kernels does not wander into explosive patterns. This is enforced by delaying adaptation until after n_0 iterations and by the ridge term ϵI_d, which keeps all eigenvalues within a controlled range.

Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	-1.1344	0.0468	1 134.3722
ν	-1.3671	0.0448	755.0555
ρ	0.6548	0.0630	706.1350

Table 3: Posterior summary for the Adaptive Metropolis sampler applied to $\boldsymbol{\theta} = (\sigma, \nu, \rho)$. The table shows posterior means, posterior standard deviations, and univariate ESS. The large ESS values show the rapid mixing under covariance adaptation and an empirical acceptance rate of 0.2173.

We see that the plots in figure 3 show the behaviour expected from an Adaptive Metropolius scheme with online covariance learning. After the burn-in at n_0, the mechanism rescales step sizes to the local posterior variance, rotates proposals along high-variance directions, and regularises small eigenmodes through ϵI_d. Consequently, the trace plots for σ, ν, ρ are stable and well-mixed after n_0, with frequent level crossings and no visible drift. Additionally, the low-lag ACF is materially reduced and decays quicly, which is reflected in the ESS figures in table 3. The distributions are approximately Gaussian on the working scale for σ and ν, while ρ concentrates away from ± 1, avoiding boundary-induced autocorrelation. Any residual autocorrelation observed is attributable to bounded support and to finite sample noise in $\hat{\Sigma}^{(n)}$ early in adaptation. Hence both effects diminish as n grows, which matches the visibly improved mixing deeper into the run.

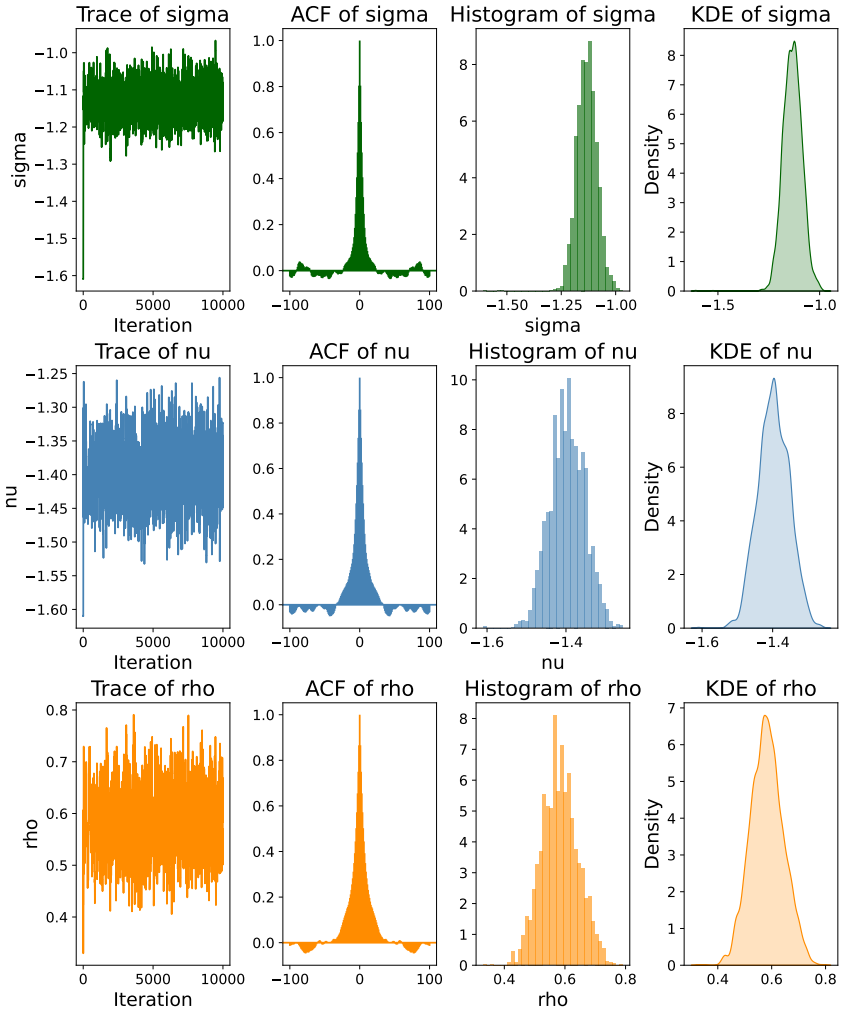


Figure 3: Adaptive MCMC diagnostic plots for each coordinate of $\boldsymbol{\theta} = (\sigma, \nu, \rho)$: (i) trace plots, (ii) autocorrelation functions (ACF), (iii) histograms, and (iv) kernel density estimates (KDE). The traces exhibit stationary exploration after burn-in with the ACFs decay sharply relative to non-adaptive baselines. The distributions are unimodal with moderate spread and ρ remains within its bounds with mass centred away from the boundaries.

2.4 Birth-Death MCMC

In option pricing with jump diffusions, the jump component of asset returns is naturally modeled as a count process such as a Poisson process to take into account the occurrence of discontinuities. A Metropolis-within-Gibbs would treat the integer-valued latent counts with a propose Gaussian random-walk which moves on the count parameter or discrete jumps of arbitrary size. The former violates the integer support and must be ad-hoc rounded, while the latter often leads to vanishing acceptance rates or poor mixing as the chain struggles to traverse the count space in small or overly large increments.

In the Merton jump-diffusion model, where the likelihood couples each latent jump count to both continuous diffusion residuals and a Poisson prior, these issues cause even more of an issue. For example, the improper proposals distort the posterior shape, and mixing stalls in regions of high latent-count probability . To address these problems, we use a *birth-death Metropolis Hastings* step within the Gibbs sampler. By proposing only local

birth or death moves, the *Birth-Death Markov Chain Monte Carlo (BDMCMC)* maintains detailed balance with the exact target posterior and achieves higher acceptance rates and more efficient exploration of the latent count space than ad-hoc discrete updates in a standard Metropolis within Gibbs framework (Green, 1995).

2.4.1 Birth-Death Process

A birth-death process is a continuous-time Markov chain $\{N(t)\}_{t\geq 0}$ taking values in the nonnegative integers $\mathbb{N}_0$, in which transitions may only increase the state by one (birth) or decrease by one (death). When the process is in state n, births occur at rate λ_n and deaths at rate μ_n, with $\mu_0 = 0$ by convention so that the chain cannot exit $\mathbb{N}_0$ downwards (think of the process as if it is a population, the population can never be negative only non-negative).Let $P_{mn}(t) = \Pr\{N(t) = n | N(0) = m\}$, where $N(t)$ is the birth-death process, then the forward-Kolmogorov equations are:

$$\frac{d}{dt}P_{mn}(t) = \lambda_{n-1}P_{m,n-1}(t) + \mu_{n+1}P_{m,n+1}(t) - (\lambda_n + \mu_n)P_{mn}(t),$$

with $\lambda_{-1} = 0$ and $\mu_0 = 0$. Equivalently, the infinitesimal generator $\mathcal{Q} = (q_{ij}$ has nonzero entries

$$q_{n,n+1} = \lambda_n, \quad q_{n,n-1} = \mu_n, \quad q_{nn} = -(\lambda_n + \mu_n),$$

and $q_{ij} = 0$ for $|i - j| > 1$. Since the total exit rate from state n is $\alpha_n = \lambda_n + \mu_n$, the waiting time in n is $\text{Exp}(\alpha_n)$. Hence, the chain jumps to $n+1$ with probability $\frac{\lambda_n}{\alpha_n}$ and to $n-1$ with probability $\frac{\mu_n}{\alpha_n}$. If the chain is positive recurrent, it admits a unique stationary distribution $\{\pi\}$ satisfying the detailed balance conditions:

$$\pi_n\lambda_n = \pi_{n+1}\mu_{n+1}, \quad \sum_{n\geq 0}\pi_n = 1.$$

Which can be solved recursively giving a solution for π_n:

$$\pi_n = \pi_0 \prod_{k=0}^{n-1}\frac{\lambda_k}{\mu_{k+1}}, \pi_0 = \left(1 + \sum_{n\geq 1}\prod_{k=0}^{n-1}\frac{\lambda_k}{\mu_{k+1}}\right)^{-1}.$$

2.4.2 Birth-Death MCMC

Similarly to section 2.2, we want to sample an unobserved count $N \in \mathbb{N}_0$ from its posterior distribution $\pi(n) \propto \ell(n)p(n)$, where $\ell(n)$ is the likelihood of the data given $N = n$, and $p(n)$ is the prior probability of $N = n$. To incorporate a birth-death Metropolis Hastings step within a larger Gibbs sampler, we suppose that the current latent state is $N^{(i^-)} = n$ at iteration i. Then, we define the proposal distribution for birth and death moves:

$$b_n = \Pr\{n \to n+1\}, \quad d_n = \Pr\{n \to n - \text{`}1\},$$

such that $b_n + d_n = 1$, $d_0 = 0$, and $b_n, d_n > 0$ for $n \geq 1$.[6] We then draw $U \sim \text{Uniform}(0, 1)$. If $U < b_n$, we set the candidate $n^* = n + 1$; otherwise $n^* = \max(n - 1, 0)$. To enforce detailed balance with respect to $\pi(\cdot)$, we compute the following acceptance probability:

$$\alpha = \frac{\pi(n^*)q(n^* \to n)}{\pi(n)q(n \to n^*)},$$

[6]We will be defining $b_n = d_n = \frac{1}{2}$ for $n \geq 1$, $b_0 = 1$.

where $q(n \to m)$ is the proposal probability of moving from state n to state m (Robert et al., 1999). It is important to note that the birth-death Metropolis-Hastings chain is irreducible whenever $b_n > 0 \ \forall n$ and $d_n > 0 \ \forall n$, and aperiodic since there is always a positive probability of rejecting a move and remaining at n. Hence, the chain converges in distribution to the target posterior $\pi(n)$.

2.4.3 Applying Birth-Death MCMC to Merton Jump Diffusion Model

Let S_t follow the Merton risk-neutral model:

$$\frac{dS_t}{S_{t^-}} = (r - \lambda k)dt + \sigma dW_t + (e^Y - 1)dN_t,$$

where N_t is a Poisson process with rate λ and jump sizes $Y \sim \mathcal{N}(\nu, \delta^2)$ meaning that $k = \mathbb{E}[e^Y - 1]$. Over a discretisation into intervals of length Δ, let $N = N_T$ be the total count.

We want to sample from the joint posterior of parameters and latent count N, but we will focus here on updating N conditional on all else. The full conditional is $\pi(n) \propto \ell(n)p(n)$, where $\ell(N)$ is the likelihood of the observed price increments $\{S_{t_{i+1}} - S_{t_i}\}|N$, and $p(N) = \mathbb{P}(N = n) = e^{-\lambda T}\frac{(\lambda T)^n}{n!}$. At iteration i, suppose the current latent count is $N^{(i)} = n$. We draw $U \sim \text{Uniform}(0, 1)$ and set the proposed count

$$n^* = \begin{cases} n + 1, & U < b_n, \\ n - 1, & U \geq b_n, \end{cases}$$

where $b_n \in (0, 1)$ is our chosen birth and $d_n = 1 - b_n$ the death probability, with $d_0 = 0$ so we never propose negative counts. Then to preserve detailed balance with respect to $\pi(n)$, we accept the move $n \to n^*$ with probability

$$\alpha = \min\left(1, \frac{\pi(n^*)q(n^* \to n)}{\pi(n)q(n \to n^*)}\right),$$

such that the proposal kernel satisfies

$$q(n \to n + 1) = b_n, \quad q(n \to n - 1) = d_n,$$

and likewise for $q(n^* \to n)$. Since

$$\frac{p(n+1)}{p(n)} = \frac{e^{-\lambda T}(\lambda T)^{n+1}/(n+1)!}{e^{-\lambda T}(\lambda T)^n/n!} = \frac{\lambda T}{n+1}, \quad \frac{p(n-1)}{p(n)} = \frac{n}{\lambda T},$$

the prior ratio is explicit. Furthermore, under the data model, conditional on $N = n$ the log-price increment over $[0, T]$ is

$$\ln S_T - \ln S_0 = \left(r - \lambda k - \frac{1}{2}\sigma^2\right)T + \sigma W_T + \sum_{i=1}^{n} Y_i,$$

so that the continuous part and the sum of n independently identically distributed normals combine into a single normal. Formally, for two neighbouring counts, $\ell(n) \sim \mathcal{N}(\mu_0 + n\nu, \sigma^2 T + n\delta^2)$, where $\mu_0 := (r - \lambda k - \frac{1}{2}\sigma^2)T$. Thus

$$\frac{\ell(n+1)}{\ell(n)} = \frac{\varphi(x; \mu_0 + (n+1)\nu, \sigma^2 T + (n+1)\delta^2)}{\varphi(x; \mu_0 + n\nu, \sigma^2 T + n\delta^2)},$$

where $\varphi(x;\mu,\sigma) = \frac{1}{\sqrt{2\pi\sigma}} \exp(-\frac{(x-\mu)^2}{2\sigma})$. Hence the birth move $n \to n+1$ is

$$\alpha_b = \min\left(1, \frac{\ell(n+1)}{\ell(n)} \frac{\lambda T}{n+1} \frac{d_{n+1}}{b_n}\right),$$

and the death move $n \to n-1$ is

$$\alpha_d = \min\left(1, \frac{\ell(n-1)}{\ell(n)} \frac{n}{\lambda T} \frac{b_{n-1}}{d_n}\right).$$

Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	-0.0092	0.0204	32.473714
λ	0.2381	0.2946	4.530036
ν	0.4606	0.1980	3.371626
δ	-0.5295	0.2700	2.495140

Table 4: Poserior summary for the Birth-Death MCMC sampler in the jump-diffusion setting, for $\boldsymbol{\theta} = (\sigma, \lambda, \nu, \delta)$ showing posterior mean, posterior standard deviation, and univariate ESS. The low ESS for λ, ν, δ relative to σ shows dependence in the jump component, consistent with strong posterior coupling between the jump and jump size law.

In figure 4 the trace plots show different behaviours across coordinates. For σ, the chain presents a frequent level crossings and relaticely short lived local persistence, showing an identified diffusion scale through the continuous component of returns. In contrast, the jump parameters display infrequent regime switches with the path remaining near a local level for extended stretches before moving to a new region. This stickiness is a structural consequence of the discrete latent count n and the acceptance probabilities for the birth/death moves. When n is near its posterior mode, the likelihood ratio disfavors moves that push the chain away from that mode. When n is far, the proposal often attempts low probability corrections. In both cases, local proposals in n pass inertia to (λ, ν, δ) through the tight coupling between the jump frequency and jump sizes. The effect is a trace with plateau like segments and delayed mean reversion for the jump parameters, which is consistent with the low ESS reported in the table.

Additionally, the ACF reinforces this picture. σ has a faster decay, whereas λ, ν, δ retain substantial correlation across lags. The marginal for σ is comparatively concentrated and near-Gaussian on the working scale. In contrast, λ is right-skewed with mass near low intensities and a diffuse tail, while ν and δ are broader and skewed, reflecting compensation between jump size location and scale. In other words, increases in ν can be partially offset by decreases in δ, and vice versa, to produce comparable aggregate jump contributions. Hence, the ridge-like coupling naturally widens the KDes and thickens tails.

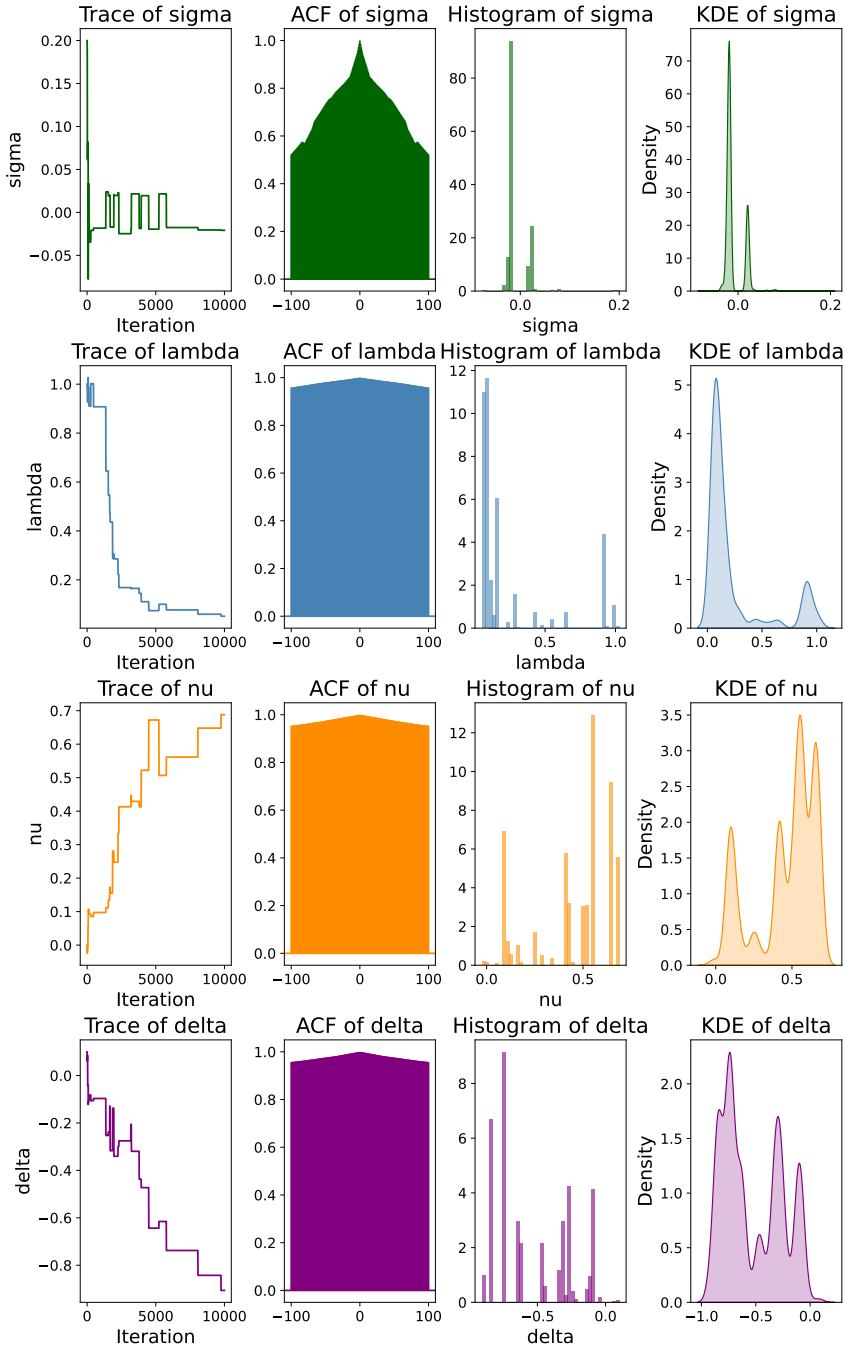


Figure 4: Diagnostic plots for the Birth-Death MCMC output by parameter $\boldsymbol{\theta} = (\sigma, \lambda, \nu, \delta)$. The traces for λ, ν, δ display long visits and slow regime changes. Their ACFs decay slowly, and the distribution are skewed with broad dispersion.

2.5 Hamiltonian Monte Carlo

Up to this point, we have struggled to get good results for complex pricing models such as Heston and Merton jump diffusion models. In many Bayesian inference problems, standard random-walk Metropolis or Gibbs samplers suffer from poor scaling in high-dimensional parameter spaces or when strong posterior correlation exist. *Hamiltonian Monte Carlo (HMC)* fixes these limitations by introducing *auxiliary momentum* variables and simulating Hamiltonian dynamics to propose moves that preserve detail balance.

The main idea is to view sampling from a target density $\pi(\boldsymbol{\theta})$ as exploring the level sets of an energy function, and to exploit gradient information to traverse parameter space efficiently (Duane et al., 1987).

Formally, let $\boldsymbol{\theta} \in \mathbb{R}^D$ denote the vector of parameters whose posterior density is $\pi(\boldsymbol{\theta}) \propto e^{-U(\boldsymbol{\theta})}$, where

$$U(\boldsymbol{\theta}) = -\log[\pi(\boldsymbol{\theta})]$$

is called the potential energy. We then introduce an auxiliary momentum variable $p \in \mathbb{R}^D$with a conditional Gaussian density $p \sim \mathcal{N}(0, M)$, where M is a positive-define *mass matrix*. The joint density over (θ, p) is then

$$\Pi(\boldsymbol{\theta}, p) = \pi(\boldsymbol{\theta})\mathcal{N}(p|0, M) \propto \exp(-H(\boldsymbol{\theta}, p)),$$

with Hamiltonian

$$H(\boldsymbol{\theta}, p) = U(\boldsymbol{\theta}) + K(p), \quad K(p) = \frac{1}{2}p^\top M^{-1}p,$$

the kinetic energy.

We can describe how $(\boldsymbol{\theta}, p)$ evolve in continuous time t through the *Hamilton's equations of motion*:

$$\frac{d\boldsymbol{\theta}}{dt} = \frac{\partial H}{\partial p} = M^{-1}p, \quad \frac{dp}{dt} = -\frac{\partial H}{\partial \boldsymbol{\theta}} = -\nabla U(\boldsymbol{\theta}). \tag{3}$$

These deterministic dynamics preserve the joint density $\Pi(\boldsymbol{\theta}, p)$, i.e, they conserve H, and are volume-preserving in phase space. Since the exact integration of (3) is generally impossible, we apply a symplectic integrator, usually *leapfrog scheme*, to approximate the flow with small step size ϵ and repeating the following three steps L times to yield a proposal $(\boldsymbol{\theta}^*, p^*)$:

$$\begin{aligned}
p\left(t + \frac{\epsilon}{2}\right) &= p(t) - \frac{\epsilon}{2}\nabla U(\boldsymbol{\theta}(t)) \qquad \text{(Half-momentum update)}, \\
\boldsymbol{\theta}(t + \epsilon) &= \boldsymbol{\theta}(t) + \epsilon M^{-1} p\left(t + \frac{\epsilon}{2}\right) \qquad \text{(Position update)}, \\
p(t + \epsilon) &= p\left(t + \frac{\epsilon}{2}\right) - \frac{\epsilon}{2}\nabla U(\boldsymbol{\theta}(t + \epsilon)) \qquad \text{Half-momentum update)}.
\end{aligned}$$

As numerical integration incurs discretisation error, the Hamiltonian is no longer exactly conversed, and to correct for this and to ensure detailed balance, we perform a Metropolis acceptance step to accept $(\boldsymbol{\theta}^*, p^*)$ with probability

$$\min\{1, \exp[-H(\boldsymbol{\theta}^*, p^*) + H(\boldsymbol{\theta}, p)]\}$$

Finally, new momentum is resampled from $\mathcal{N}(0, M)$ before commencing the next trajectory. HMC can propose moves that travel in parameter space with high probability by using gradient information, reducing the random-walk behaviour that simpler MCMC methods use.

2.5.1 Applying HMC to the Heston Model

Recall that the Heston model under the risk-neutral measure is

$$\begin{aligned}
dS_t &= \mu S_t dt + S_t\sqrt{v_t}dW^{\mathbb{Q}}_{t,1}, \\
dv_t &= \kappa(\xi - v_t)dt + \sigma\sqrt{v_t}dW^{\mathbb{Q}}_{t,2}, \\
d\langle W^{\mathbb{Q}}_{t,1}, W^{\mathbb{Q}}_{t,2}\rangle &= \rho dt.
\end{aligned}$$

For simplicity, we will write $W^{\mathbb{Q}}_{t,i} := W_{t,i}$ for $i = 1, 2$. To apply HMC to Heston, we require defining a suitable potential energy function and computing its gradient with respect to the model parameters $\boldsymbol{\theta} = (\kappa, \xi, \sigma, \rho)$ and the latent volatility path $\{v_t\}$. Given discrete observations $S_{t_0}, S_{t_1}, ..., S_{t_N}$, we can write the joint posterior as

$$\pi(\boldsymbol{\theta}, \{v_i\}|\{S_i\}) \propto \pi(\boldsymbol{\theta})\pi(v_0) \prod_{i=1}^{N} \ell(S_i|S_{i-1}, v_{i-1}, \boldsymbol{\theta})\ell(v_i|v_{i-1}, \boldsymbol{\theta}).$$

Similarly to section 2.2, we can rewrite the problem under Euler discretisation as:

$$\mathcal{L}(S_i|S_{i-1}, v_{i-1}, \boldsymbol{\theta}) := \log \ell(S_i|S_{i-1}, v_{i-1}, \boldsymbol{\theta}) = -\frac{1}{2}\left[\log(2\pi v_{i-1}S^2_{i-1}\Delta t) + \frac{(\Delta \log S_i - (r - \frac{1}{2}v_{i-1}\Delta t)^2}{v_{i-1}\Delta t}\right],$$
$$\mathcal{L}(v_i|v_{i-1}, \boldsymbol{\theta}) := \log \ell(v_i|v_{i-1}, \boldsymbol{\theta}) = -\frac{1}{2}\left[\log(2\pi\sigma^2 v_{i-1}\Delta t) + \frac{(\Delta v_i - \kappa(\xi - v_{i-1})\Delta t)^2}{\sigma^2 v_{i-1}\Delta t}\right],$$

where $\Delta \log S_i = \log(\frac{S_i}{S_{i-1}})$ and $\Delta v_i = v_i - v_{i-1}$. Thus the potential energy is

$$U(\boldsymbol{\theta}.\{v_i\}) = -\log \pi(\boldsymbol{\theta}) - \log \pi(v_0) - \sum_{i=1}^{N} \left[\mathcal{L}(S_i|S_{i-1}, v_{i-1}, \boldsymbol{\theta}) + \mathcal{L}(v_i|v_{i-1}, \boldsymbol{\theta})\right].$$

Gradient components are then obtained by differentiating U with respect to each parameter and each latent v_i. For example,

$$\frac{\partial U}{\partial v_{i-1}} = -\frac{\partial}{\partial v_{i-1}}\mathcal{L}(S_i|S_{i-1}, v_{i-1}, \boldsymbol{\theta}) - \frac{\partial}{\partial v_{i-1}}\mathcal{L}(v_i|v_{i-1}, \boldsymbol{\theta}) - \frac{\partial}{\partial v_{i-1}}\mathcal{L}(v_{i-1}|v_{i-2}, \boldsymbol{\theta}),$$

each term makes closed form expressions involving polynomials and inverse functions of v_{i-1}. We then assemble these gradients into the vector $\nabla_{\boldsymbol{\theta},v}U$ and implement the leapfrog steps in the joint space of $\boldsymbol{\theta}$ and $\{v_i\}$, tuning ϵ, M, and L.

HMC Heston: Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
μ	0.1995	0.2109	184.222600
κ	1.5888	0.7302	362.592367
ξ	0.2174	0.1135	220.770254
σ	1.0890	0.1922	91.640614
ρ	-0.5650	0.1991	50.603245

Table 5: Posterior summary for the Heston parameters under Hamiltonian Monte Carlo. The table shows posterior means, posterior standard deviations, and univariate ESS for $\boldsymbol{\theta} = (\mu, \kappa, \xi, \sigma, \rho)$. The ESS values show efficient exploration for κ and ξ, moderate efficiency for μ, and comparatively slower mixing for σ and ρ, reflecting residual stiffness and boundary effects in those coordinates.

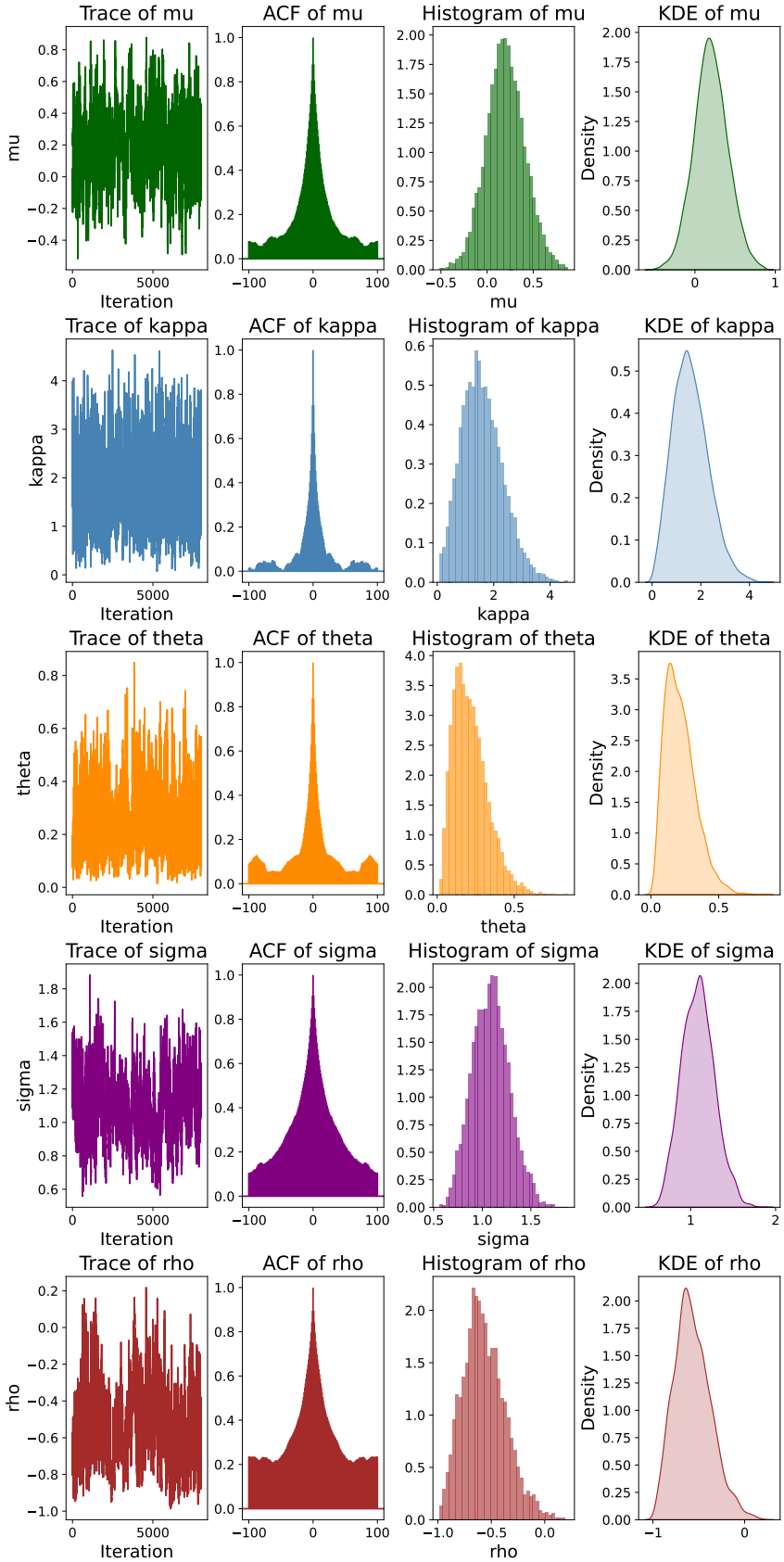


Figure 5: HMC diagnostic plots for the Heston model by parameter $\boldsymbol{\theta} = (\mu, \kappa, \xi, \sigma, \rho)$: (i) trace plots, (ii) autocorrelation functions, (iii) histograms, and (iv) kernel density estimates. The traces are stationary with frequent level crossings and the ACFs for κ and ξ decay rapidly. μ shows moderate persistence, whereas σ and ρ exhibit slower ACF decay. Distributions are unimodal with moderate spread for μ, κ, ξ. The σ marginal is broader and ρ concentrates on negative values and remains away from the boundaries.

Figure 5 shows reduced random-walk behavior, stationary traces, and largely unimodal marginals, which is expected with HMC. The mean reversion (κ) and long-run variance (ξ) show the cleanest diagnostics of short memory traces and sharp decaying ACFs with tight KDEs. This is reflected in the ESS results in table 5. The μ coordinate mixes adequately but with visibly slower ACF decay than κ and ξ. In contrast the volatility of volatility (σ) and leverage (ρ) retain longer memory. σ has a broader marginal and a noticeable ACF tail, while ρ concentrates on negative values with slower ACF decay.

Gradient-informed proposals allow long, high-acceptance trajectories along the principal directions of (κ, ξ), mitigating the ridge structure that hampers random walk samplers. Hence their rapiud ACF decay and larger ESS. The σ parameter couples to both the latent variance dynamics and the scaling of the price model, producing a stiffer curvature and wider marginal. The finite step size and a generic mass matrix leavesw some residual autocorrelation and hence lowering ESS. For $\rho \in (-1, 1)$, the parameterisation induces geometric compression as $|\rho| \to 1$, so equal momentum updates translate into smaller moves near the boundaries. In other words, the unconstrained Fisher-z scale $\theta_5 = \arctan(\rho)$, then

$$\frac{d\rho}{d\theta_5} = 1 - \rho^2,$$

where $\boldsymbol{\theta}_{unc} := (\theta_1, \theta_2, \theta_3, \theta_4, \theta_5)$ are the unconstrained parameters of the model. So a fixed size update un θ_5 induces a change in ρ of order $(1 - \rho^2)\Delta\theta_5$ as $|\rho| \to 1$ this factor vanishes, geometrically compressing at the boundaries.

2.5.2 Applying HMC to the Multi-Asset Black-Scholes Model

Recall that the multi-asset Black-Scholes model under the risk neutral measure is

$$\begin{aligned} dS_{1,t} &= rS_{1,t}dt + \sigma S_{1,t}dW_{1,t}, \\ dS_{2,t} &= rS_{2,t}dt + \nu S_{2,t}dW_{2.t}, \\ d\langle W_1, W_2\rangle_t &= \rho dt. \end{aligned}$$

To apply HMC to multi-asset Black-Scholes, we require defining a suitable potential energy function and computing its gradient with respect to the model parameters $\boldsymbol{\theta} = (\sigma, \nu, \rho)$. Suppose observations are available at times $0 = t_0 < t_1 < ... < t_N$ with uniform step $\Delta t = t_i - t_{i-1}$. Define the bivariate log-return increment $\Delta X_i := (\Delta X_{1,i}, \Delta X_{2,i})^\top$ with $\Delta X_{1,i} = X_{1,i} - X_{1,i-1} = \log(\frac{S_{1,t_i}}{S_{1,t_{i-1}}})$ and $\Delta X_{2,i} = X_{2,i} - X_{2,i-1} = \log(\frac{S_{2,t_i}}{S_{2,t_{i-1}}})$. Similarly to section 2.2, we can use the Euler-Maruyama approximation to get Gaussian increments with mean

$$\boldsymbol{\mu}(\boldsymbol{\theta}) = \left(\left(r - \frac{1}{2}\sigma^2\right)\Delta t, \left(r - \frac{1}{2}\nu^2\right)\Delta t\right)^\top,$$

and covariance

$$\Sigma(\boldsymbol{\theta}) = \Delta t \begin{pmatrix} \sigma^2 & \rho\sigma\nu \\ \rho\sigma\nu & \nu^2 \end{pmatrix},$$

where $\boldsymbol{\theta} := (\sigma, \nu, \rho)$. Writing residuals as $\boldsymbol{z}_i(\boldsymbol{\theta}) := \Delta\boldsymbol{X}_i - \boldsymbol{\mu}(\boldsymbol{\theta})$, then each increment likelihood can be written as the bivariate normal density

$$\ell(\Delta\boldsymbol{X}_i|\boldsymbol{\theta}) = \frac{1}{2\pi\sqrt{\det\Sigma(\boldsymbol{\theta})}} \exp\left(-\frac{1}{2}\boldsymbol{z}_i(\boldsymbol{\theta})^\top\Sigma(\boldsymbol{\theta})^{-1}\boldsymbol{z}_i(\boldsymbol{\theta})\right).$$

Hence the log likelihood for the entire series is

$$\mathcal{L}(\boldsymbol{\theta}) = \sum_{i=1}^{N} \log \ell(\Delta \boldsymbol{X}_i | \boldsymbol{\theta}) = -\frac{N}{2} \log((2\pi)^2) - \frac{N}{2} \log \det \Sigma(\boldsymbol{\theta}) - \frac{1}{2} \sum_{i=1}^{N} \boldsymbol{z}_i(\boldsymbol{\theta})^\top \Sigma(\boldsymbol{\theta})^{-1}.$$

Let $\pi(\boldsymbol{\theta})$ denote a prior density. The potential energy for HMC is the negative log-posterior:

$$U(\boldsymbol{\theta}) = -\log \pi(\boldsymbol{\theta}) - \mathcal{L}(\boldsymbol{\theta}),$$

and the HMC force is $\nabla_{\boldsymbol{\theta}} U(\boldsymbol{\theta}) = -\nabla_{\boldsymbol{\theta}} \log \pi(\boldsymbol{\theta}) - \nabla_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$. Recall that:

$$\det \Sigma(\boldsymbol{\theta}) = \Delta t^2 \sigma^2 \nu^2 (1-\rho^2), \quad \Sigma(\boldsymbol{\theta})^{-1} = \frac{1}{\Delta t(1-\rho^2)} \begin{pmatrix} \sigma^{-2} & -\rho(\sigma\nu)^{-1} \\ -\rho(\sigma\nu)^{-1} & \nu^{-2}. \end{pmatrix}$$

We now need to differentiating $\nabla_{\boldsymbol{\theta}} \mathcal{L}$, to begin we need to differentiate the following terms as shown below

$$\begin{aligned} \frac{\partial}{\partial \vartheta}(\boldsymbol{z}^\top \Sigma^{-1} \boldsymbol{z}) &= -2(\partial_\vartheta \boldsymbol{\mu})^\top \Sigma^{-1} \boldsymbol{z} - \boldsymbol{z}^\top \Sigma^{-1} (\partial_\vartheta \Sigma) \Sigma^{-1} \boldsymbol{z}, \\ \frac{\partial}{\partial \vartheta} \log \det \Sigma &= \operatorname{tr}(\Sigma^{-1} \partial_\vartheta \Sigma), \end{aligned} \tag{4}$$

for $\vartheta \in \{\sigma, \nu, \rho\}$. The derivatives of the mean and covariance for each parameter is

$$\partial_\vartheta \boldsymbol{\mu} = (-\sigma \Delta t, 0)^\top, \quad \partial_\nu \boldsymbol{\mu} = (0, -\nu \Delta t)^\top, \quad \partial_\rho \boldsymbol{\mu} = (0,0)^\top, \tag{5}$$

and

$$\partial_\sigma \Sigma = \Delta t \begin{pmatrix} 2\sigma & \rho\nu \\ \rho\nu & 0 \end{pmatrix}, \quad \partial_\nu \Sigma = \Delta t \begin{pmatrix} 0 & \rho\sigma \\ \rho\sigma & 2\nu \end{pmatrix}, \quad \partial_\rho \Sigma = \Delta t \begin{pmatrix} 0 & \sigma\nu \\ \sigma\nu & 0 \end{pmatrix}. \tag{6}$$

From $\det \Sigma$ we can see that

$$\partial_\sigma \log \det \Sigma = \frac{2}{\sigma}, \quad \partial_\nu \log \det \Sigma = \frac{2}{\nu}, \quad \partial_\rho \log \det \Sigma = -\frac{2\rho}{1-\rho^2}. \tag{7}$$

Substituting (4), (5), (6), and (7) into $\partial_{\boldsymbol{\theta}} \mathcal{L}(\boldsymbol{\theta})$ we get, for each $\vartheta \in \{\sigma, \nu, \rho\}$,

$$\partial_\vartheta \mathcal{L}(\boldsymbol{\theta}) = -\frac{N}{2} \partial_\vartheta \log \det \Sigma(\boldsymbol{\theta}) + \sum_{i=1}^{N} (\partial_\vartheta \boldsymbol{\mu})^\top \Sigma(\boldsymbol{\theta})^{-1} \boldsymbol{z}_i(\boldsymbol{\theta}) + \frac{1}{2} \sum_{i=1}^{N} \boldsymbol{z}_i(\boldsymbol{\theta})^\top \Sigma(\boldsymbol{\theta})^{-1} (\partial_\vartheta \Sigma(\boldsymbol{\theta})) \Sigma(\boldsymbol{\theta})^{-1} \boldsymbol{z}_i(\boldsymbol{\theta}).$$

For numerical stability and constraint handling, we will work on an unconstrained parameter vector $\boldsymbol{\theta}_{unc} = (\theta_1, \theta_2, \theta_3) \in \mathbb{R}^3$, like previously, via the smooth bijection $\sigma = \exp(\theta_1), \nu = \exp(\theta_2)$, and $\rho = \tanh(\theta_3)$. If priors are directly on $\boldsymbol{\theta}_{unc}$, then the log-Jacobian does not enter the acceptance ratio, and the chain rule gives

$$\frac{\partial U}{\partial \theta_1} = \frac{\partial U}{\partial \sigma} \sigma, \quad \frac{\partial U}{\partial \theta_2} = \frac{\partial U}{\partial \nu} \nu, \quad \frac{\partial U}{\partial \theta_3} = \frac{\partial U}{\partial \rho} (1-\rho^2).$$

However, if the priors are specified on (σ, ν, ρ), then U on the $\boldsymbol{\theta}_{unc}$ space receives the additional log-Jacobian term $\log(\sigma\nu(1-\rho^2))$. Differentiating this contributes to $+1$ to ∂_{θ_1} and to ∂_{θ_2} and -2ρ to ∂_{θ_3}. Then we apply the above chain-rule multipliers.

We then introduce a momentum $p \sim \mathcal{N}(0, M)$ with positive definite mass matrix M defines the Hamiltonian $H(\boldsymbol{\theta}_{unc}, p) = U(\boldsymbol{\theta}_{unc}) + \frac{1}{2}p^\top M^{-1}p$. Hamilton's equations are then

$$\frac{d\boldsymbol{\theta}_{unc}}{dt} = M^{-1}p, \quad \frac{dp}{dt} = -\nabla_{\boldsymbol{\theta}_{unc}} U(\boldsymbol{\theta}_{unc}).$$

The leapfrog integrator with step size $\epsilon > 0$ and L steps implements a half step momentum update using $-\nabla_{\boldsymbol{\theta}_{unc}} U$, a full-step position update via $M^{-1}p$, and a final half-step momentum update. The the proposal $(\boldsymbol{\theta}^*_{unc}, q)$ is accepted with probability

$$\alpha = \min\left\{1, \exp(-H(\boldsymbol{\theta}^*_{unc}, q) + H(\boldsymbol{\theta}_{unc}, p)\right\}.$$

HMC Multi Black-Scholes: Summary Statistics

Parameter	Mean	Standard Deviation	Effective Sample Size (ESS)
σ	0.3205	0.0144	5161.588590
ν	0.2460	0.0110	5378.332800
ρ	0.5189	0.0459	5195.554648

Table 6: Posterior summary for the Multi Black-Scholes parameters under Hamiltonian Monte Carlo. The table shows posterior means, posterior standard deviations, and univariate ESS for $\boldsymbol{\theta} = (\sigma, \nu, \rho)$. The ESS values indicate highly efficient exploration and very weak correlation in the draws, with tight marginal uncertainty for σ and ν and moderate dispersion for ρ.

The diagnostic plots in figure 6 show the efficiently tuned Hamiltonian Monte Carlo. Trajectories traverse level sets of the posterior with minimal random walk behaviour, creating stationary traces with dense coverage of the support and ACFs that fall to zero after a few lags. The distributions of σ and ν are near-Gaussian, consistent with the small posterior standard deviations in table 6, and ρ exhibits a wider but still unimodal marginal with the trace remaining comfortably away from the boundaries, which is reflected in a slightly broader distribution yet comparably fast ACF decay.

When we compare these results to the results in section 2.3, HMC attains significantly higher ESS and visibly faster ACF decay for each coordinate. In figure 3, the ACFs decay more slowly and the traces show longer plateaus, particularly for the correlation coordinate, whereas the HMC's momentum-based proposals generate moves aligned with the posterior curvature, suppressing diffusive behaviour and reducing integrated autocorrelation time. In addition, the leapfrog integrator preserves a constant Hamiltonian along trajectories, delivering high acceptance without shrinking step sizes, whereas the adaptive random walk in figure 3 must balance acceptance against step length, leading to shorter effective moves in high-curvature directions. Hence, for the same nominal iteration budget, HMC produces larger ESS with narrower distributions for all parameters.

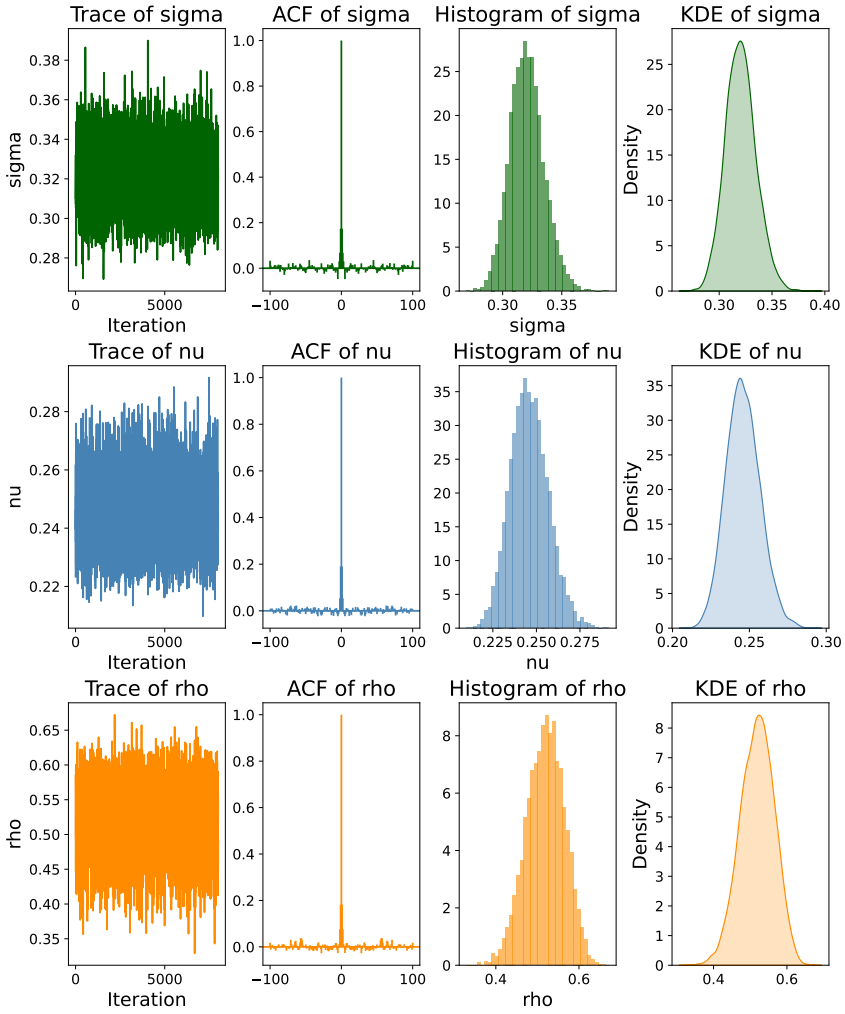


Figure 6: HMC Diagnostic plots for the multi-asset Black-Scholes model by parameter $\boldsymbol{\theta} = (\sigma, \nu, \rho)$: (i) trace plots, (ii) autocorrelation functions, (iii) histograms, and (iv) kernel density estimates. Traces show stationary exploration with frequent level-crossing and ACFs decay to near-zero at very low lags. Distributions are unimodal and concentrated for σ and ν, and moderately broader for ρ.

3 Exotic Options

3.1 Asian options

Asian options are a type of single asset exotic option where the payoff function is defined in terms of the average price of the underlying, instead of the final price. We will be considering Geometric Asian options in this paper. The payoff function of this option is:

$$\Phi = \max(A_T - K, 0)$$

$$\textit{where } A_T = \exp\Big\{\frac{1}{T}\int_0^T \log(S_t)\,dt\Big\}$$

3.2 Margrabe's Option & Rainbow Options

3.3 Pricing Under Black-Scholes

3.3.1 Asian

Recall from section TODO the dynamics of the log prices in a Black-Scholes pricing model under risk neutral measure $\mathbb{Q}$:

$$dX_t = (r - \frac{1}{2}\sigma^2)dt\ + \sigma dW_t$$

Integrating both sides between 0 and t, we obtain

$$\int_0^t dX_t = \int_0^t (r - \frac{1}{2}\sigma^2)dt + \sigma\int_0^t dW_t$$

$$X_t - X_0 = (r - \frac{1}{2}\sigma^2)t + \sigma dW_t$$

$$\log(S_t) = \log(S_0) + (r - \frac{1}{2}\sigma^2)t + \sigma dW_t$$

Now, substituting this expression for $\log(S_t)$ into our expression for A_T:

$$A_T = \exp\Big\{\frac{1}{T}\int_0^T \log(S_t)\,dt\Big\} = \exp\Big\{\frac{1}{T}\int_0^T \log(S_0) + (r - \frac{1}{2}\sigma^2)t + \sigma W_t\,dt\Big\}$$

Considering only the exponentiated term:

$$\textit{let } Y = \frac{1}{T}\int_0^T \log(S_0) + (r - \frac{1}{2}\sigma^2)t + \sigma W_t\,dt$$

$$Y = \log(S_0) + \frac{T}{2}(r - \frac{1}{2}\sigma^2) + \frac{\sigma}{T}\int_0^T W_t dt$$

As W_t is a stochastic process, this integral will not have a deterministic solution. However, due to properties of Brownian motion, we know this integral is normally distributed with mean 0. So consider the variance of this distribution:

$$\mathbb{V}ar\Big[\int_0^T W_t dt\Big] = \mathbb{E}\Big[\Big(\int_0^T W_t dt\Big)^2\Big]$$

$$\mathbb{V}ar\Big[\int_0^T W_t dt\Big] = \mathbb{E}\Big[\Big(\int_0^T W_t dt\Big)\Big(\int_0^T W_t dt\Big)\Big] = \mathbb{E}\Big[\Big(\int_0^T W_t dt\Big)\Big(\int_0^T W_s ds\Big)\Big]$$

Now rewriting this integral as a double integral:

$$\mathbb{V}ar\Big[\int_0^T W_t dt\Big] = \mathbb{E}\Big[\int_0^T \int_0^T W_s W_t ds dt\Big]$$

$$= \int_0^T \int_0^T \mathbb{E}[W_s W_t] ds dt = \int_0^T \int_0^T \min(t,s) ds dt$$

Now, dividing this $[0,T] \times [0,T]$ region into 2 parts, one with $s \leq t$, and the other with $s > t$

$$\mathbb{V}ar\Big[\int_0^T W_t dt\Big] = \int\int_{s \leq t} s\, ds dt + \int\int_{s>t} t\, ds dt$$

Now substituting in bounds for these regions:

$$\mathbb{V}ar\Big[\int_0^T W_t dt\Big] = \int_0^T \int_0^t s\, ds dt + \int_0^T \int_0^s t\, ds dt = 2\int_0^T \int_0^t s\, ds dt = 2 \cdot \frac{T^3}{6} = \frac{T^3}{3}$$

Thus $\mathbb{V}ar\Big[\frac{\sigma}{T}\int_0^T W_t dt\Big] = \frac{\sigma^2}{T^2} \cdot \frac{T^3}{3} = \frac{\sigma^2 T}{3}$.
Therefore, we can see that $\int_0^T W_t \sim N(0, \frac{T^3}{3})$, so using properties of the normal distribution:

$$Y \sim N(\hat{\mu}, \nu^2),$$

Where:

- $\hat{\mu} = \log(S_0) + \frac{T}{2}(r - \frac{1}{2}\sigma^2)$
- $\nu = \sigma\sqrt{\frac{T}{3}}$

Now we can rewrite the expression for A_T as:

$$A_T = e^Y$$

From this we can see that A_T is log-normally distributed, so the price of the option will take a similar form to that of Black-Scholes for a European call option.
Defining the standard normal variable as Z:

$$Z = \frac{Y - \hat{\mu}}{\nu} \implies Y = \nu Z + \hat{\mu}$$

$$\mathbb{Q}(A_T > K) = \mathbb{Q}(\log(A_T) > \log(K))$$

$$= \mathbb{Q}(\nu Z + \hat{\mu} > \log(K)) = \mathbb{Q}\Big(Z > \frac{\log(K) - \hat{\mu}}{\nu}\Big)$$

$$\mathbb{Q}(A_t > K) = 1 - \mathbb{Q}\Big(Z < \frac{\log(K) - \hat{\mu}}{\nu}\Big) = 1 - \Phi\Big(-\frac{\log(K) - \hat{\mu}}{\nu}\Big)$$

As d_2 is the risk neutral probability that the option finishes in the money:

$$\mathbb{Q}(A_T > K) = \mathbb{Q}(Z > -d_2) = 1 - \mathbb{Q}(Z < d_2)$$

From this it can be seen that:

$$d_2 = \frac{\hat{\mu} - \log(K)}{\nu}$$

Just as with European call options, $d_2 = d_1 - \nu \implies d_1 = d_2 + \nu$ So:

$$d_1 = \nu + d_2 = \frac{\hat{\mu} + \nu^2 - \log(K)}{\nu}$$

Thus we have all of the expressions required for the final option price.

$$C = e^{-rT}(S_0 e^{\hat{\mu}+\frac{1}{2}\nu^2}\Phi(d_1) - K\Phi(d_2))$$

Where:

- $\hat{\mu} = \log(S_0) + \frac{T}{2}(r - \frac{1}{2}\sigma^2)$
- $\nu = \frac{\sigma^2 T}{3}$
- $d_1 = \frac{\hat{\mu}+\nu^2-\log(K)}{\nu}$
- $d_2 = \frac{\hat{\mu}-\log(K)}{\nu}$

3.3.2 Margrabe's

Margrabe's two-asset exchange option is an option which one holds the right to exchange one unit of asset 2 for one unit of asset 1 at time T. Formally speaking, the payoff function is

$$\Phi(S_{1,T}, S_{2,T}, T) = \max(S_{1,T} - S_{2,T}), 0). \tag{8}$$

To price this claim we introduce the log-ratio process

$$X_t = \ln S_{1,t} - \ln S_{2,t}. \tag{9}$$

Recall that under risk-neutral measure $\mathbb{Q}$, each asset price $S_{i,t}$ in the Black-Scholes framework satisfies

$$d \ln S_{i,t} = \left(r - \frac{1}{2}\sigma_i^2\right) dt + \sigma_i dW_{i,t},$$

for $i = 1, 2$, where r is the risk-free rate, σ_i, the volatility of asset i, and the Brownian motions satisfy $d\langle W_{t,1}, W_{t,2}\rangle = \rho dt$. By applying Itô's lemma to (9) and linearity of stochastic integrals,

$$dX_t = d \ln S_{1,t} - d \ln S_{2.t} = \left[\left(r - \frac{1}{2}\sigma_1^2\right) - \left(r - \frac{1}{2}\sigma_2^2\right)\right] dt + \sigma_1 dW_{1,t} - \sigma_2 dW_{2,t}$$

. Since $d\langle W_{t,1}, W_{t,2}\rangle = \rho dt$, we cam define a single Brownian motion W_t and an effective volatility

$$\sigma_{\text{eff}} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2},$$

so that

$$\sigma_1 dW_{1,t} - \sigma_2 dW_{2,t} = \sigma_{\text{eff}} dW_t.$$

Hence, we can write dX_t as

$$dX_t = -\frac{1}{2}(\sigma_1^2 - \sigma_2^2)dt + \sigma_{\mathbf{eff}} dW_t.$$

Integrating both sides from 0 to T gives

$$X_T - X_0 = -\frac{1}{2}(\sigma_1^2 - \sigma_2^2)T + \sigma_{\text{eff}} \int_0^T dW_t$$

$$\Rightarrow X_T = X_0 - \frac{1}{2}(\sigma_1^2 - \sigma_2^2)T + \sigma_{\text{eff}} W_T,$$

and therefore X_T is normally distributed with mean

$$\mu_X = X_0 - \frac{1}{2}(\sigma_1^2 - \sigma_2^2)T,$$

and variance

$$\mathbb{V}\text{ar}[X_T] = \sigma_{\text{eff}}^2 T.$$

The exchange-option payoff can be written as

$$(S_{1,T} - S_{2,T})^+ = S_{2,T}(\exp(X_T) - 1)^+.$$

Its time-zero fair price is the discounted risk-neutral expectation:

$$C = e^{-rT}\, \mathbb{E}_{\mathbb{Q}} \left[S_{2,T}(\exp(X_T) - 1)^+ \right].$$

Since under $\mathbb{Q}^{(2)}$ the drift of X_t becomes $-\frac{1}{2}\sigma_{\text{eff}}^2$, the law od X_T under this measure is

$$X_T \sim \mathcal{N}\left(X_0 - \frac{1}{2}\sigma_{\text{eff}}^2 T, \sigma_{\text{eff}}^2 T \right).$$

Hence, if we let $F_t = \frac{S_{1,t}}{S_{2,t}} = e^{X_t}$, then F_t is log-normal, and we can define

$$d_1 = \frac{X_0 + \frac{1}{2}\sigma_{\text{eff}}^2 T}{\sigma_{\text{eff}}\sqrt{T}}, \quad d_2 = d_1 - \sigma_{\text{eff}}\sqrt{T},$$

where $X_0 = \ln \frac{S_{1,0}}{S_{2,0}}$. The expectaion of $(F_T - 1)^+$ under the log-normal law is then

$$\mathbb{E}_{\mathbb{Q}^{(2)}} \left[(F_T - 1)^+ \right] = N(d_1) - N(d_2),$$

where N is the standard normal cumulative distribution. Hence the Margrabe formula in closed form is

$$C = S_{1,0} N(d_1) - S_{2,0} N(d_2). \tag{10}$$

3.4 Pricing Under Merton Jump Diffusion

3.4.1 Asian

To begin, we denote N_t as a Poisson process with rate λ, and by $\{Y_i\}_{i \geq 1}$ as an independent identically distributed sequence of jump sizes with

$$Y_i \sim \mathcal{N}(\nu, \delta^2),$$

and define the compensator $k = \mathbb{E}[e^Y - 1] = e^{\nu + \frac{1}{2}\delta^2} - 1$. Then the risk-neutral SDE is

$$\frac{dS_t}{S_{t^-}} = (r - \lambda k)dt + \sigma dW_t + (e^Y - 1)dN_t.$$

By using Itô's lemma, the log-price $X_t = \ln S_t$ satisfies

$$dX_t = (r - \lambda k - \frac{1}{2}\sigma^2)dt + \sigma dW_t + Y dN_t,$$

such that

$$X_t = \ln S_0 \left(r - \lambda k - \frac{1}{2}\sigma^2 \right) t + \sigma W_t + \sum_{i=1}^{N_t} Y_i.$$

Since we are interested in the geometric Asian payoff

$$A_T = \exp\left\{ \frac{1}{T} \int_0^T X_s ds \right\},$$

we will set

$$X_T := \frac{1}{T} \int_0^T X_s ds, \tag{11}$$

so that $A_T = e^{X_T}$. By substituting (11) into Y_t, we get

$$\begin{aligned} X_T &= \frac{1}{T} \int_0^T \left[X_0 + (r - \lambda k - \frac{1}{2}\sigma^2)s + \sigma W_s + \sum_{i=1}^{N_s} Y_i \right] ds \\ &= \frac{1}{T} \int_0^T \ln S_0 ds + \frac{1}{T} \int_0^T (r - \lambda k - \frac{1}{2}\sigma^2) s ds + \frac{1}{T} \int_0^T \sigma W_s ds + \frac{1}{T} \int_0^T \sum_{i=1}^{N_s} Y_i ds \\ &= \ln S_0 + \left(r - \lambda k - \frac{1}{2}\sigma^2 \right) \frac{T}{2} + \frac{1}{T} \int_0^T \sigma W_s ds + \frac{1}{T} \int_0^T \sum_{i=1}^{N_s} Y_i ds. \end{aligned}$$

To handle $\frac{\sigma}{T} \int_0^T W_s ds$, we must find the mean and variance. However, we know $\int_0^T W_s ds$ is Gaussian with mean zero and variance

$$\mathbb{V}\text{ar}\left[\int_0^T W_s ds \right] = \int_0^T \int_0^T \mathbb{E}[W_u W_v] du dv = \int_0^T \int_0^T \min(u, v) du dv = \frac{T^3}{3}.$$

Hence

$$\frac{\sigma}{T} \int_0^T W_s ds \sim \mathcal{N}\left(0, \frac{\sigma^2 T}{3} \right).$$

The jump process term can be solved using Fubini. We first note that:

$$\sum_{i=1}^{N_s} Y_i = \sum_{i=1}^{N_T} Y_i \mathbb{1}_{\tau_i \leq s},$$

where τ_i are the jump times up to T. Hence

$$\frac{1}{T} \int_0^T \sum_{i=1}^{N_s} Y_i ds = \frac{1}{T} \sum_{i=1}^{N_T} \int_0^T Y_i \mathbb{1}_{\tau_i \leq s} ds = \frac{1}{T} \sum_{i=1}^{N_T} Y_i (T - \tau_i).$$

Therefore

$$X_T = m_0 + \frac{\sigma}{T} \int_0^T W_s ds + J_T,$$

where

$$m_0 := \ln S_0 + \left(r - \lambda k - \frac{1}{2}\sigma^2\right)\frac{T}{2}, \qquad J_T := \frac{1}{T}\sum_{i=1}^{N_T} Y_i(T - \tau_i).$$

Conditioned $N_T = n$, the jump times $\{\tau_i\}$ are independent identically distributed uniform on $[0, T]$. Hence

$$\mathbb{E}[T - \tau_i | N_T = n] = \frac{T}{2},$$

and $\mathbb{E}[Y_i] = \nu$. Thus

$$\mathbb{E}[J_t | N_T = n] = \frac{1}{T} n\nu \frac{T}{2} = \frac{n\nu}{2}.$$

By averaging over $N_T \sim \text{Pois}(\lambda T)$, we get

$$\mathbb{E}[J_T] = \sum_{n=0}^{\infty} \frac{n\nu}{2} e^{-\lambda T} \frac{(\lambda T)^n}{n!} = \frac{\nu}{2}\,\mathbb{E}[N_T] = \frac{\nu}{2}\lambda T.$$

To find the variance, we first note that conditional on $N_T = n$ and jump times, the variance of the sum is

$$\mathbb{V}\text{ar}\left[\frac{1}{T}\sum_{i=1}^{N_T} Y_i(T - \tau_i)\right] = \frac{1}{T^2}\sum_{i=1}^{n} \mathbb{V}\text{ar}[Y_i(T - \tau_i)],$$

since the $Y_i(T - \tau_i)$ are independent given the τ_i. Now,

$$\mathbb{V}\text{ar}[Y_i(T - \tau_i)] = \mathbb{E}[Y_i^2]\,\mathbb{E}[(T - \tau_i)^2] - (\mathbb{E}[Y_i])^2(\mathbb{E}[T - \tau_i])^2.$$

From generating functions, we have $\mathbb{E}[Y_i^2] = \nu^2 + \delta^2$, $\mathbb{E}[T - \tau_i] = \frac{T}{2}$, and

$$\mathbb{E}[(T - \tau_i)^2] = \int_0^T \frac{(T - u)^2}{T} du = \frac{T^2}{3}.$$

Hence

$$\mathbb{V}\text{ar}[Y_i(T - \tau_i)] = (\nu^2 + \delta^2)\frac{T^2}{3} - \nu^2\frac{T^2}{4} = \frac{T^2}{12}(4\nu^2 + 4\delta^2 - 3\nu^2) = \frac{T^2}{12}(\nu^2 + 4\delta^2).$$

Therefore, conditional on $N_T = n$,

$$\mathbb{V}\text{ar}[J_T | N_t = n] = \frac{n}{T^2}\frac{T^2}{12}(\nu^2 + 4\delta^2) = \frac{n}{12}(\nu^2 + 4\delta^2)$$

By using the property $\mathbb{V}\text{ar}[J_t] = \mathbb{E}[\mathbb{V}\text{ar}(J_t | N_T)] + \mathbb{V}\text{ar}(\mathbb{E}[J_T | N_T])$, and $\mathbb{E}[N_T] = \lambda T$ and $\mathbb{V}\text{ar}[N_T] = \lambda T$,

$$\begin{aligned}\mathbb{V}\text{ar}[J_T] &= \frac{\lambda T}{12}(\nu^2 + 4\delta^2) + \mathbb{V}\text{ar}\left(\frac{N_t\nu}{2}\right) = \frac{\lambda T}{12}(\nu^2 + 4\delta^2) + \left(\frac{\nu}{2}\right)^2 \lambda T \\ &= \lambda T\left[\frac{\nu^2 + 4\delta^2}{12} + \frac{\nu^2}{4}\right] = \frac{\lambda T}{3}(\nu^2 + \delta^2).\end{aligned}$$

By plugging back into (11), we can approximate X_T by a Gaussian random variable matching its first two moments with

$$\mu_X = X_0 + \left(r - \lambda k - \frac{1}{2}\sigma^2\right)\frac{T}{2} + \frac{\lambda\nu T}{2},$$

and variance

$$\sigma_X^2 = \frac{\sigma^2 T}{3} + \frac{\lambda T}{3}(\nu^2 + \delta^2).$$

Since A_T is log normal with these parameters, the time zero price of the geometric Asain call, with strike K, is

$$C = e^{-rT}\,\mathbb{E}[(A_T - K)^+] = e^{-rT}[e^{\mu_X + \frac{1}{2}\sigma_X^2} N(d_1) - KN(d_2)],$$

where N is the standard normal cumulative distribution and

$$d_1 = \frac{\mu_X + \sigma_X^2 - \ln K}{\sigma_Y}, \quad d_2 = d_1 - \sigma_Y.$$

3.4.2 Margrabe's

Recall from subsection 3.3.2 that $X_t = \ln S_{1,t} - \ln S_{2,t}$, where

$$\frac{dS_{i,t}}{S_{i,t^-}} = (r - \lambda_i k_i)dt + \sigma_i dW_{i,t} + (e^{Y_i} - 1)dN_{i,t},$$

$$d\langle W_1, W_2\rangle = \rho dt, \quad i = 1, 2.$$

We can apply Itô's lemma to get

$$dX_t = [r - \lambda_1 k_1 - \frac{1}{2}\sigma_1^2]dt - [r - \lambda_2 k_2 - \frac{1}{2}\sigma_2^2]dt + \sigma_1 dW_{1,t} - \sigma_2 dW_{2,t} + Y_1 dN_{1,t} - Y_2 dN_{2,t}.$$

We can rewrite drift as

$$\mu_0 := [r - \lambda_1 k_1 - \frac{1}{2}\sigma_1^2] - [r - \lambda_2 k_2 - \frac{1}{2}\sigma_2^2] = -\frac{1}{2}[\sigma_1^2 - \sigma_2^2] - [\lambda_1 k_1 - \lambda_2 k_2],$$

and similarly in section 3.3.2, we can combine the Brownian terms into a single Brownian motion W_t with effective volatility:

$$\sigma_{\text{eff}} = \sqrt{\sigma_1^2 + \sigma_2^2 - 2\rho\sigma_1\sigma_2},$$

hence giving

$$dX_t = \mu_0 dt + \sigma_{\text{eff}} dW_t + Y_1 dN_{1,t} - Y_2 dN_{2,t}.$$

Integrating both sides then gives:

$$\begin{aligned} X_T - X_0 &= \int_0^T \mu_0 dt + \int_0^T \sigma_{\text{eff}} dW_t + \sum_{i=1}^{N_{1,T}} Y_{1,i} - \sum_{j=1}^{N_{2,T}} Y_{2,j}, \\ &= \mu_0 T + \sigma_{\text{eff}} W_T + \sum_{i=1}^{N_{1,T}} Y_{1,i} - \sum_{j=1}^{N_{2,T}} Y_{2,j}. \end{aligned}$$

Recall that the Margrabe payoff is $\max(S_{1,T} - S_{2,T}, 0) = S_{2,T}(e^{X_T} - 1)^+$. Under the risk neutral measure $\mathbb{Q}_2$ with S_2 as numéraire, the discount ratio $F_t = \frac{S_{1,t}}{S_{2,t}} = e^{X_t}$ is a martingale.[7] Thus the time-zero price is

$$\Pi = S_{2,0}\,\mathbb{E}_{\mathbb{Q}^{(2)}}\left[\left(e^{X_T} - 1\right)^+\right]. \tag{12}$$

[7]The proof is straight forward since $e^{-rt}S_{1,t}$ and $e^{-rt}S_{2,t}$ are both martingales.

We can now condition on the realised jump counts $N_{1,T} = n_1$ and $N_{2,T} = n_2$. Since we are switching from $\mathbb{Q}$ to $\mathbb{Q}^{(2)}$, the Brownian motion part shifts, the jump intensity changes, and the jump-size law of asset 2 gets exponentially titled. More precisely,

$$\tilde{\lambda}_2 = \lambda_2 \, \mathbb{E}_{\mathbb{Q}}[e^{Y_2}] = \lambda_2 e^{\nu_2 + \frac{1}{2}\delta_2^2} = \lambda_2(1 + k_2),$$

and the titled jump size $\tilde{Y}_2$ has density proportional to $e^y f_{Y_2}(y)$ so for normal jumps we have

$$\tilde{Y}_2 \sim \mathcal{N}(\nu_2 + \delta_2^2, \delta_2^2).$$

Hence the asset 2 jump contributions under $\mathbb{Q}^{(2)}$ shifts to

$$\sum_{j=1}^{\tilde{N}_{2,T}} \tilde{Y}_{2,j}, \qquad \tilde{N}_{2,T} \sim \text{Pois}(\tilde{\lambda}_2 T).$$

Additionally, under $\mathbb{Q}^{(2)}$, the log-ratio drift becomes $\mu_* = -\frac{1}{2}\sigma_{\text{eff}}^2 - \lambda_1 k_1 - \tilde{\lambda}_2 \tilde{k}_2$, where $\tilde{k}_2 := \mathbb{E}_{\mathbb{Q}^{(2)}}[\exp(-\tilde{Y}_2) - 1] = \exp(-\nu_2 - \frac{1}{2}\delta_2^2) - 1$. Since

$$\tilde{\lambda}_2 \tilde{k}_2 = \lambda_2 e^{\nu_2 + \frac{1}{2}\delta_2^"} \left(e^{-\nu_2 - \frac{1}{2}\delta_2^2} - 1 \right) = -\lambda_2 k_2,$$

we can write $\mu_* = -\frac{1}{2}\sigma_{\text{eff}}^2 - \lambda_1 k_1 + \lambda_2 k_2$.

We can now condition on the realised jump counts $N_{1,T} = n_1$ and $\tilde{N}_{2,T} = n_2$. Since $N_{1,T}$, $\tilde{N}_{2,T}$ are independent Poisson random variables with parameters $\lambda_1 T$ and $\tilde{\lambda}_2 T$ respectively, we can write

$$\mathbb{E}_{\mathbb{Q}^{(2)}} \left[\left(e^{X_T} - 1 \right)^+ \right] = \sum_{n_1=0}^{\infty} \sum_{n_2=0}^{\infty} E_{\mathbb{Q}^{(2)}} \left[\left(e^{X_T} - 1 \right)^+ | N_{1,T} = n_1, N_{2,T} = n_2 \right] \mathbb{P}(N_{1,T} = n_1) \mathbb{P}(N_{2,T} = n_2).$$

For fixed (n_1, n_2), the conditional distribution of

$$X_T = X_0 + \mu_* T + \sigma_{\text{eff}} W_t + \sum_{i=1}^{n_1} Y_{1,i} - \sum_{j=1}^{n_2} \tilde{Y}_{2,j},$$

will have a normal distribution of:

$$X_T \sim \mathcal{N}(X_0 + \mu_* T + n_1 \nu_1 - n_2(\nu_2 + \delta_2^2), \quad \sigma_{\text{eff}}^2 T + n_1 \delta_1^2 + n_2 \delta_2^2).$$

This is due to the two sums of independent normals are themselves normal with means $n_1\nu_1$ and $n_2\nu_2$, and variance $n_1\delta_1^2$ and $n_2\delta_2^2$ respectively. Under this conditional normal law, the expectation of $(e^{X_T} - 1)^+$ is exactly the Black-Scholes-style one-asset call formula applied to the log-normal variable e^{X_T}.Define

$$d_1(n_1, n_2) = \frac{X_0 + \mu_* T + n_1 \nu_1 - n_2(\nu_2 + \delta_2^2) + \sigma_{\text{eff}}^2 T + n_1 \delta_1^2 + n_2 \delta_2^2}{\sqrt{\sigma_{\text{eff}}^2 T + n_1 \delta_1^2 + n_2 \delta_2^2}},$$

$$d_2(n_1, n_2) = d_1(n_1, n_2) - \sqrt{\sigma_{\text{eff}}^2 T + n_1 \delta_1^2 + n_2 \delta_2^2}.$$

For simplicity we will define the conditional distribution of X_T as $X_T \sim \mathcal{N}(m, v)$. We can express the expectation of (12) with the density of X_T:

$$\begin{aligned}\mathbb{E}[(e^{X_T}-1)^+] &= \int_0^\infty (e^x-1)f(x)dx = \int_0^\infty e^x f(x)dx - \int_0^\infty f(x)dx \\ &= \int_0^\infty \frac{1}{\sqrt{2\pi v}} \exp\left(x - \frac{(x-m)^2}{2v}\right) dx - \mathbb{P}[X_T > 0] \\ &= \int_0^\infty \frac{1}{\sqrt{2\pi v}} \exp\left(x - \frac{(x-m)^2}{2v}\right) dx - \Phi\left(\frac{m}{\sqrt{v}}\right),\end{aligned}$$

where Φ is the standard normal cumulative density function. By completing the square we can rearrange the exponential term in the first integral as

$$e^{m+\frac{1}{2}v} \int_0^\infty \frac{1}{\sqrt{2\pi v}} \exp\left(-\frac{(x-(m+v))^2}{2v}\right) dx,$$

hence, by a change in variable of $z = \frac{x-(m+v)}{\sqrt{v}}$, we get the following:

$$e^{m+\frac{1}{2}v} \int_{-(m+v)/\sqrt{v}}^\infty \frac{1}{\sqrt{2\pi}} e^{-\frac{z^2}{2}} dz = e^{m+\frac{1}{2}v}\Phi\left(\frac{m+v}{\sqrt{v}}\right) = e^{m+\frac{1}{2}v}\Phi(d_1),$$

where $d_1 = \frac{m+v}{\sqrt{v}}$. If we take $d_2 = \frac{m}{\sqrt{v}}$, then we can write the expectation from (12) as

$$\mathbb{E}[(e^{X_T}-1)^+] = \int_0^\infty (e^x-1)f(x)dx = e^{m+\frac{1}{2}v}\Phi(d_1) - \Phi(d_2).$$

Hence,

$$\mathbb{E}_{\mathbb{Q}_2}[(e^{X_T}-1)^+] = e^{m+\frac{1}{2}v}\Phi(d_1) - \Phi(d_2).$$

and

$$\mathbb{E}_{\mathbb{Q}_2}[(e^{X_T}-1)^+|n_1,n_2] = e^{m_{n_1,n_2}+\frac{1}{2}\nu_{n_1,n_2}}\Phi(d_1(n_1,n_2)) - \Phi(d_2(n_1,n_2)).$$

Putting everything together we get:

$$\Pi = S_{2,0} \sum_{n_1=0}^\infty \sum_{n_2=0}^\infty [\Phi(d_1(n_1,n_2)) - \Phi(d_2(n_1,n_2))] \frac{e^{-\lambda_1 T}(\lambda_1 T)^{n_1}}{n_1!} \frac{e^{-\lambda_2 T}(\lambda_2 T)^{n_2}}{n_2!}.$$

3.5 Pricing Under Heston Model

3.5.1 Asian

Recall under risk-neutral measure $\mathbb{Q}$, the Heston model is

$$\begin{aligned} dS_t &= \mu S_t dt + S_t\sqrt{v_t} dW_{t,1}^{\mathbb{Q}}, \\ dv_t &= \kappa(\xi - v_t)dt + \sigma\sqrt{v_t} dW_{t,2}^{\mathbb{Q}}, \\ d\langle W_{t,1}^{\mathbb{Q}}, W_{t,2}^{\mathbb{Q}}\rangle &= \rho dt. \end{aligned}$$

Similarly as before using Itô's lemma,

$$dX_t = \left(r - \frac{1}{2}v_t\right)dt + \sqrt{v_t} dW_{t,1}.$$

If the geometric Asian payoff is

$$A_T = \exp\left(\frac{1}{T}\int_0^T X_s ds\right)$$

with the fair call price

$$\Pi = e^{-rT}\,\mathbb{E}_{\mathbb{Q}}[(A_T - K)^+],$$

then to get a closed-form expression, we must first calculate the moment generating function of $\frac{1}{T}\int_0^T X_s ds$, i.e., $M(u) = \mathbb{E}[\exp(\frac{1}{T}\int_0^T X_s ds)]$. If we consider the conditional expectation

$$F(t,x,v) = \mathbb{E}\left[\exp\left(u\frac{1}{T}\int_t^T X_s ds\right)\middle| X_t = x, v_t = v\right],$$

then over the next small interval $[t, t+\Delta t]$, we earn the factor $\exp(\frac{u}{T}\int_t^{t+\Delta t} X_s ds) \approx \exp(\frac{u}{T}x\Delta t)$. Hence we are in the new state $(X_{t+\Delta t}, v_{t+\Delta t})$ from which the remaining expected payoff is $F(t+\Delta t, X_{t+\Delta t}, v_{t+\Delta t})$:

$$F(t,x,v) = \mathbb{E}\left[e^{\frac{u}{T}x\Delta t}F(t+\Delta t, X_{t+\Delta t}, v_{t+\Delta t})\middle| X_t = x, v_t = v\right]. \tag{13}$$

Additionally, if we let $\mathcal{L}$ be the infinitesimal generator of the 2D diffusion (X_t, v_t), then the generator acting on a twice-differentiable test function $f(x,v)$ is

$$\mathcal{L}f = \left(r - \frac{1}{2}v\right)\partial_x f + \kappa(\xi - v)\partial_v f + \frac{1}{2}v\partial_{xx}f + \rho\sigma v\partial_{xv}f + \frac{1}{2}\sigma^2 v\partial_{vv}f.$$

If we then expand $F(t+\Delta t, X_{t+\Delta t}, v_{t+\Delta t})$ to first order in Δt using Itô's Lemma and the generator $\mathcal{L}$ of the (X_t, v_t) process, we get a time expansion of $F(t+\Delta t, X_{t+\Delta t}, v_{t+\Delta t}) = F(t,x,v) + \partial_t F\Delta t + o(\Delta t)$ and a space expansion via the continuous diffusion part:

$$\mathcal{L}F = \frac{1}{2}v\partial_{xx}F + \rho\sigma v\partial_{xv}F + \frac{1}{2}\sigma^2 v\partial_{vv}F + \left(r - \frac{1}{2}v\right)\partial_x F + \kappa(\xi - v),$$

hence by Itô,

$$\mathbb{E}[F(t+\Delta t, X_{t+\Delta t}, v_{t+\Delta t})] = F(t,x,v) + (\partial_t F + \mathcal{L}F)\Delta t + o(\Delta t). \tag{14}$$

We can expand $e^{(u/T)x\Delta t}$ to get $e^{(u/T)x\Delta t} = 1 + \frac{u}{T}x\Delta t + \wr(\Delta t)$. By substituting this and (14) into (13) we get:

$$\begin{aligned} F(t,x,v) &= \mathbb{E}\left[(1 + \frac{u}{T}x\Delta t)(F + (\partial_t F + \mathcal{L}F)\Delta t)\right] + o(\Delta t) \\ &= F + \left[\partial_t F + \mathcal{L}F + \frac{u}{T}xF\right]\Delta t + o(\Delta t). \\ \implies \partial_t F + \mathcal{L}F + \frac{u}{T}xF &= 0. \end{aligned}$$

By multiplying by -1 we get the backward form equation:

$$\partial_t F = \mathcal{L}F - \frac{u}{T}xF.$$

Substituting in the generator terms we get

$$-\partial_t F = \frac{1}{2}v\partial_{xx}F + \rho\sigma v\partial_{xv}F + \frac{1}{2}\sigma^2 v\partial_{vv}F + (r - \frac{1}{2}v)\partial_x F + \kappa(\xi - v)\partial_v F - \frac{u}{T}xF,$$

with terminal condition $F(T,x,v)=1$.[8] We now take the exponential-affine form in the variables x and v as

$$F(t,x,v)=\exp\left(A(\tau)+B(\tau)x+C(\tau)v\right),$$

where $\tau=T-t$, and A,B,C are scalar functions of τ only (Heston, 1993). Computing the derivatives we get:

$$\begin{aligned}
\partial_t F &= -\frac{dA}{d\tau}F-\frac{dB}{d\tau}xF-\frac{dC}{d\tau}vF,\\
\partial_x F &= BF,\quad \partial_{xx}F=B^2F,\\
\partial_v F &= CF,\quad \partial_{vv}F=C^2F,\quad \partial_{xv}F=BCF.
\end{aligned}$$

Substitute into the PDE and cancelling out F:

$$\frac{dA}{d\tau}+\frac{dB}{d\tau}x+\frac{dC}{d\tau}v=\frac{1}{2}vB^2+\rho\sigma vBC+\frac{1}{2}\sigma^2vC^2+\left(r-\frac{1}{2}v\right)B+\kappa(\xi-v)C-\frac{u}{T}x.$$

Collecting terms:

$$\begin{aligned}
&\frac{dB}{d\tau}=-\frac{u}{T},\quad B(0)=0\quad(\text{since } F(T,\cdot)=1),\\
&\implies\quad B(\tau)=-\frac{u}{T}\tau,\\
&\frac{dC}{d\tau}(\tau)=\frac{1}{2}(B(\tau))^2+\rho\sigma B(\tau)C(\tau)+\frac{1}{2}\sigma^2C(\tau)^2-\frac{1}{2}B(\tau)-\kappa C(\tau),\quad C(0)=0,\\
&\frac{dA}{d\tau}(\tau)=rB(\tau)+\kappa\xi C(\tau),\quad A(0)=0.
\end{aligned}$$

[9]Using the solutions to A,B,C up to $\tau=T$, we have at $t=0$:

$$M(u)=F(0,X_0,v_0)=\exp(A(T)+B(T)X_0+C(T)v_0).$$

In particular $M(1)=\mathbb{E}[e^{\frac{1}{T}\int_0^T X_s ds}]$ and $M(0)=1$. The geometric-Asian call price is then

$$\Pi=e^{-rT}\left[M(1)-KM(0)\right]=e^{-rT}\left[e^{A(T)+B(T)X_0+C(T)v_0}-K\right].$$

3.5.2 Margrabe's

Under the original risk neutral measure $\mathbb{Q}$, we define each asset price $S_{i,t}$ and its variance $v_{i,t}$ as

$$\begin{aligned}
dS_{i,t}&=rS_{i,t}dt+\sqrt{v_{i,t}}S_{i,t}dW_{i,t},\\
dv_{i,t}&=\kappa_i(\xi_i-v_{i,t})dt+\sigma_i\sqrt{v_{i,t}}dZ_{i,t},\\
d\langle W_1,W_2\rangle_t&=\rho_S dt,\quad d\langle W_i,Z_i\rangle_t=\rho_i dt,
\end{aligned}$$

and all other covariations zero, and $i=1,2$. By using Itô's lemma on $X_t=\ln S_{1,t}-\ln S_{2,t}$,

$$dX_t=\left(r-\frac{1}{2}v_{1,t}\right)dt-\left(r-\frac{1}{2}v_{2,t}\right)dt+\sqrt{v_{1,t}}dW_{1,t}-\sqrt{v_{2,t}}dW_{2,t}.$$

[8]This is formally known as the *Feynman-Kac PDEs* (Oksendal, 2013), where F satisfies the backward PDE on $[0,T]$

[9]The solution to these PDEs is in the Appendix.

Grouping the variance and combining the two Brownian increments into a single dW_t,

$$\Sigma_t^2 = v_{1,t} + v_{2,t} - 2\rho_s\sqrt{v_{1,t}v_{2,t}},$$

$$\implies dX_t = -\frac{1}{2}(v_{1,t} - v_{2,t})dt + \Sigma_t dW_t.$$

We want to switch to the S_2 numéraire measure $\mathbb{Q}_2$, under which the discounted ratio $F_t = e^{X_t}$ must be a martingale. Applying Itô to e^{X_t} and setting its continuous dt term to zero forces $-\frac{1}{2}(v_{1,t} - v_{2,t}) = -\frac{1}{2}\Sigma_t^2$. Thus under $\mathbb{Q}^{(2)}$,

$$dX_t = -\frac{1}{2}\Sigma_t^2 dt + \Sigma_t dW_t.$$

Integrating from 0 to T:

$$X_T - X_0 = -\frac{1}{2}\int_0^T \Sigma_t^2 dt + \int_0^T \Sigma_t dW_t.$$

Let $I := \int_0^T \Sigma_t^2 dt$, then given the entire path $\{\Sigma\}_t$, the stochastic integral $\int_0^T \Sigma_t dW_t$ is normal with mean 0 and variance I, and hence, conditionally $X_T|\{\Sigma\}_t \sim \mathcal{N}(X_0 - \frac{1}{2}I, I)$. Like previously, if we write the margrabe payoff as $(E^{X_T} - 1)^+$, its conditional expectation is exactly the Black-Scholes exchange formula with log-mean $m = X_0 - \frac{1}{2}I$ and varianec $\nu = I$. Hence, similarly to previous subsections:

$$\mathbb{E}\left[\left(e^{X_T} - 1\right)\middle|\, \Sigma\right] = \Phi\left(\frac{m+\nu}{\sqrt{\nu}}\right) - \Phi\left(\frac{m}{\sqrt{\nu}}\right),$$

where $d_1(I) = \frac{X_0 + \frac{1}{2}I}{\sqrt{I}}$ and $d_2(I) = \frac{X_0 - \frac{1}{2}I}{\sqrt{I}}$. If we let $g(I) = \Phi\left(\frac{X_0+\frac{1}{2}I}{\sqrt{I}}\right) - \Phi\left(\frac{X_0-\frac{1}{2}I}{\sqrt{I}}\right)$, and I is non-negative random variable, then by the law of the unconscious statistician,

$$\mathbb{E}[g(I)] = \int_0^\infty g(j) f_I(j) dj,$$

where $f_I(i)$ is the unknown density of I. Rather than work with f_I directly, we will use its characteristic function

$$\varphi(z) = \mathbb{E}[e^{izI}] = \int_0^\infty e^{izj} f_I(j) dj$$

which can be written in closed form due to the affine structure of the Heston variance (Duffie et al., 2000). The identity that links expectations of $g(I)$ to $\varphi(z)$ is the *Fourier-inversion formula*. Formally, if we define the Fourier transform of g by

$$\tilde{g}(z) = \int_0^\infty e^{izj} g(j) dj,$$

the Fubini's theorem gives

$$\int_0^\infty g(j) f_I(j) dj = \int_0^\infty g(j)\left[\frac{1}{2\pi}\int_{-\infty}^\infty e^{-izj}\varphi(z)dz\right]dj = \frac{1}{2\pi}\int_{-\infty}^\infty \varphi(z)\tilde{g}(-z)dz.$$

Since $g(j)$ vanishes for $j < 0$ and is real-valued, we can reduce this to an integral over $z \in [0, \infty]$ of the real part:

$$\mathbb{E}[g(I)] = \frac{1}{2\pi}\int_{-\infty}^\infty \varphi(z)\tilde{g}(-z)dz.$$

We can evaluate this integral using the fact that g is real valued, so $\tilde{g}(-z) = \overline{\tilde{g}(-z)}$, and $\varphi(-z) = \overline{\varphi(z)}$, giving us

$$\int_{-\infty}^{\infty} \varphi(z)\tilde{g}(-z)dz = \int_{-\infty}^{0} \varphi(z)\tilde{g}(-z)dz + \int_{0}^{\infty} \varphi(z)\tilde{g}(-z)dz.$$

Let $u = -z$, which means that z goes from $-\infty$ to 0, u goes from $+\infty$ to 0 and $dz = -du$. Thus

$$\int_{-\infty}^{0} \varphi(z)\tilde{g}(-z)dz = \int_{\infty}^{0} \varphi(-u)\tilde{g}(u)(-du) = \int_{0}^{\infty} \varphi(-u)\tilde{g}(u)du.$$

Since g is real-valued, its transform satisfies $\tilde{g}(u) = \overline{\tilde{g}(-u)}$, and the characteristic function of a real-valued random variable satisfies $\varphi(-u) = \overline{\varphi(u)}$, hence $\varphi(-u)\tilde{g}(u) = \overline{\varphi(u)}\ \overline{\tilde{g}(-u)} = \overline{\varphi(u)\tilde{g}(-u)}$. Thus giving:

$$\int_{-\infty}^{\infty} \varphi(z)\tilde{g}(-z)dz = \int_{0}^{\infty} \varphi(z)\tilde{g}(-z)dz + \int_{0}^{\infty} \overline{\varphi(z)\tilde{g}(-z)}dz = 2\int_{0}^{\infty} \Re[\varphi(z)\tilde{g}(-z)]dz.$$

Hence

$$\mathbb{E}[g(I)] = \frac{1}{2\pi} \cdot 2\Re\left\{\int_{0}^{\infty} \varphi(z)\tilde{g}(-z)dz\right\} = \frac{1}{\pi}\int_{0}^{\infty} \Re[\varphi(z)\tilde{g}(-z)]dz$$

(Lewis, 2001) showed that this converts into

$$\mathbb{E}[g(I)] = g(\infty) - \frac{1}{\pi}\int_{0}^{\infty} \Re[\varphi(z)\hat{g}(-z)]dz,$$

where $g(\infty) = \lim_{i\to\infty} g(i) = 1$, since as $i \to \infty$ the option is almost surely in the money. Additionally, $\hat{g}(z)$ is a slightly modified transform of g, explicitly computable in terms of $e^{\frac{1}{2}iz} - 1$ and the factor $\frac{1}{iz}$ that arises from integrating by parts to handle the payoff kink at $i = 0$.

Finally, this gives us a fair price for the Margrabe option of

$$\Pi = S_{2,0}\left[1 - \frac{1}{\pi}\int_{0}^{\infty} \Re\left[\varphi(z)\frac{e^{\frac{1}{2}iz} - 1}{iz}e^{-izX_0}\right]dz\right].$$

4 Results

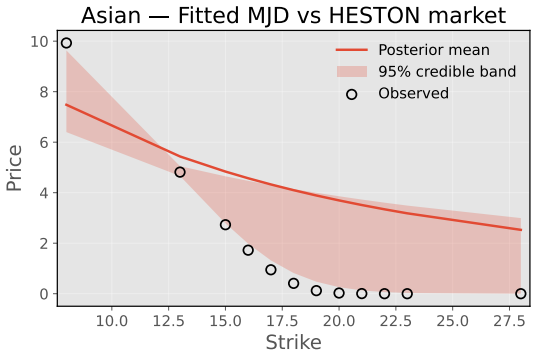
Up to this point we have built and demonstrated different Markov Chain Monte Carlo techniques on different option pricing models. We started with a random walk Metropolis on a standard Black-Scholes model. We then introduced stochastic volatility as a hidden Markov chain with the Heston model and tackled this problem using a Metropolis-within-Gibbs. We then looked at a multi-asset model using Adaptive Metropolis to tackle the issue of poor exploration in the proposal space. In this section we test the Markov Chain Monte Carlo. When looking at the Merton Jump Diffusion model, we handled the volatility jump discrete random variable by using a Birth-Death process, before looking at Hamiltonian Monte Carlo across multi-assets and a single-asset Heston model. Finally, we solved the analytical solutions for the these pricing models. In this section, we will look at a financial simulation of exotic options in the corn and soybean markets. In particular the Asian and Margrabe options. We will sample daily adjusted closes for CORN and SOYB from Yahoo Finance over the time period 1st June 2022 to 1st June 2025. We will feed their log-price series into our MCMC calibrations, and a continuously-compounded risk-free rate at $r = 1\%$. To build maturities and strikes to listed markets, we collect the CORN/SOYB option chain and set the Asian panel to the furthest available CORN call expiry with its native strike grid, and build the Margrabe panel on a tensor of 10 exchange-ratio strikes $K \in [0.5, 1.2]$ and the first four positive CORN expiries, expressed as year fractions from the valuation date. Finally, we compare the models to a benchmark model which will use benchmark parameters for the respective model. For MJD, Black-Scholes, and Heston, we use the $(0.2, 1.0, 0.0, 0.1)$, $(0.3, 0.25, 0.0)$, and $(0.04, 1.5, 0.04, 0.5, -0.4)$ respectively.

4.1 Non-HMC Samplers

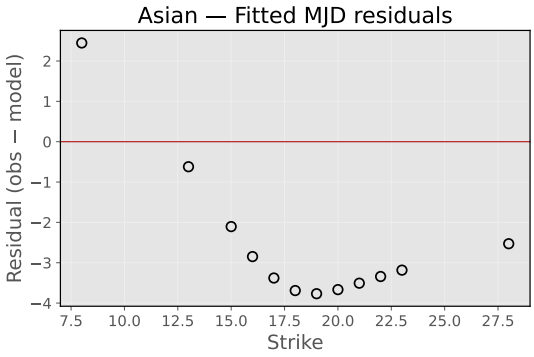
For calibration we start off with the non-HMC samplers and then run HMC samplers. Specifically, for the two-asset Black-Scholes (σ, ν, ρ) we run an Adaptive Random Walk Metropolis on the transformed parameter space $(\log \sigma, \log \nu, \operatorname{arctanh} \rho)$ for 20,000 iterations with online covariance adaption, starting at 500, cooling of 0.995, and a diagonal initial proposal. We then transform draws back to the natural scale and apply a 20% burn-in. These posterior draws feed the Margrabe pricer and uncertainty bands.

For the single-asset Merton jump-diffusion on CORN $(\sigma, \lambda, \nu, \delta)$ we use a Birth-Death MCMC inside a Metropolis-within-Gibbs with 20,000 iterations. The latent jump counts are initialised from a Poisson prior at the daily intensity, and accepted moves jointly update parameters and jump counts. Posterior samples are converted from daily to annualised units for pricing. We additionally added an extra 2% white noise to both Asian and Margrabe synthetic quotes.

4.1.1 Asian Option under Merton Jump Diffusion Model



(a) Fitted Asian option prices (Merton Jump Diffusion model) vs strike, with 95% Bayesian credible interval (red band) and observed market prices (black circles). The MJD calibration closely follows the observed price curve, especially near the at-the-money region.



(b) Residuals for the MJD calibration. The horizontal red line at 0 describes perfect calibration. The residuals (black circles) are overestimation below the line and underestimation above the line.

Figure 7: Fitted Asian option prices against strike (a) and Residuals for the MJD calibration (b).

The calibrated BDMCMC sampler generated an Asian option pricing function that would price options slightly higher than the market observed prices. Figure 7a show that the posterior mean price (red line) as a function of strike captures the dynamic observed price levels (black circles) across strikes, although predicting these values higher in the out-of-the-money strikes. The 95% credible band around the mean is relatively wide in the non-central strike ranges suggesting the model parameters are not well-constrained there, and all observed prices fall slightly outside it until a strike value of 20.0 or higher. This shows that the mixing of the Birth Death MCMC did not fix the data well and that the Bayesian calibration's uncertainty is inconsistent with the data's noise level. The band's broadening for extreme strikes reflects its uncertainty in pricing in very deep in-the-money (ITM) or out-of-the-money (OTM) options. Intuitively, these extrapolated regions are less informed by data and more sensitive to model parameter variability. The price cure is smooth and exhibits the expected monotonic decrease with strike. This model mispricing is also seen when looking at the residuals of the fitted model in figure 7b. The residuals scatter far from zero for most strikes, further evidencing the poor quality fit. The figure shows a smile-shaped pattern between the strikes 12.5 to 27.5, suggesting an overestimated option value for these prices. This again stems back to poor variable selection for the MJD model. This difference was at its highest around the mid-range OTM strikes and the difference was at its lowest around the deep-ITM option values. This miscalibration was also seen in figure 4 earlier showing the poor mixing.

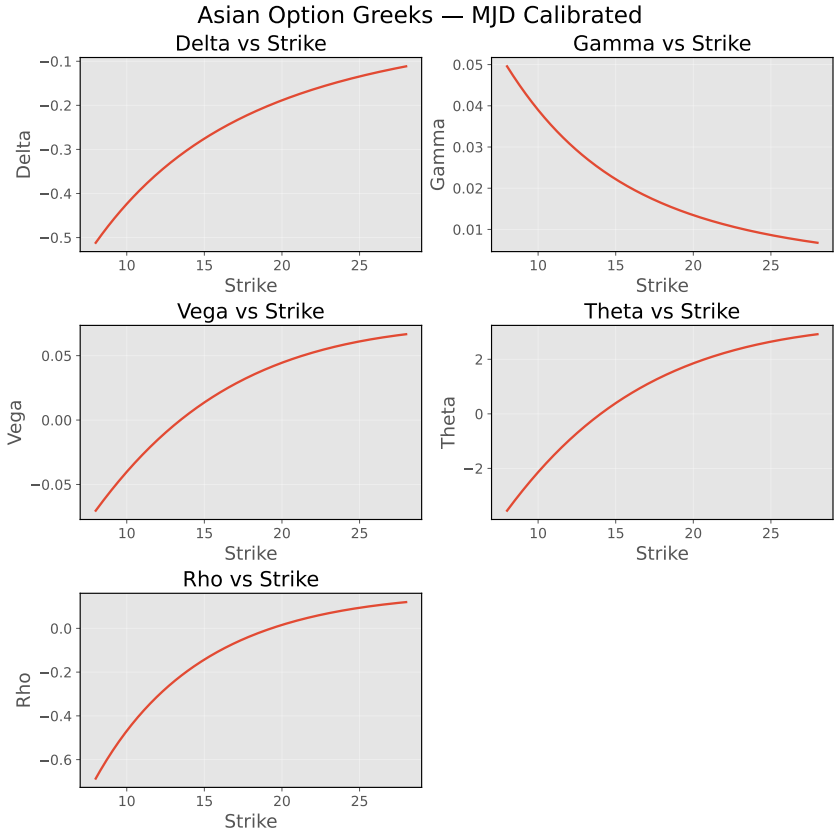
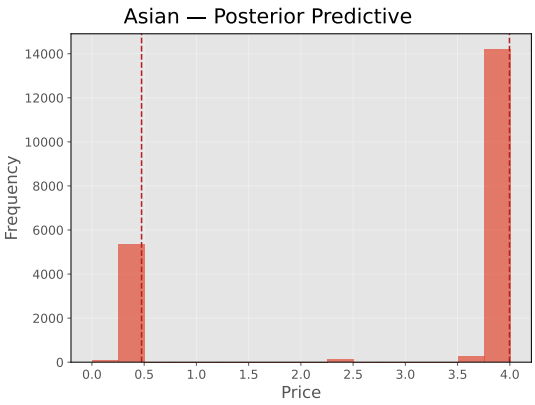
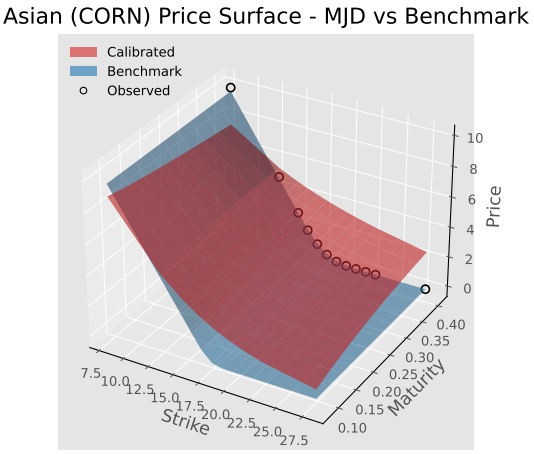


Figure 8: Greeks for the Asian option under the calibrated MJD model, as functions of strike. Delta vs Strike: Delta inclines as strike increases, ranging roughly from -0.5 at low strikes to a -0.1 at high strikes. Gamma vs Strike: Gamma peaks at the extreme OTM region and approaches zero at ITM extremes. Vega vs Strike: Vega is highest at intermediate strikes and diminishes from deep ITM or OTM strikes. Theta vs Strike: Theta is most pronounced around the ITM regions. Rho vs Strikes: Rho in this setting decreases with strike. All Greeks are computed via the posterior mean parameters of the MJD model.

Figure8 shows the Greeks of the Asian option across strikes, presenting the five primary Greeks under the MJD model calibration. The Delta curve slopes upwards as strike increases. For OTM strikes, Delta is at its lowest, whereas for high strikes Delta is lower. Quantitatively, in the MJD model the deep OTM Delta is slightly below the Black-Scholes limit of 1.0 in magnitude, which is consistent with jump diffusion theory that even an OTM call has a chance to be knocked ITM by a large downward jump (Tunaru et al., 2017). Hence, the option's sensitivity to the underlying is tempered. Looking at the Gamma, we observe that Gamma is at its highest in the OTM calls and falls off to near zero for deep ITM strikes. This reflects that the Delta of the option changes most rapidly when the option is near the OTM. This is does not align with option theory, which states that Gamma is highest ATM and slopes to zero at deep OTM and ITM options. Vega, Theta, and Rho all share similar charts with their lowest values at the OTM call options and their highest values at ITM calls. This again breaks the fundamentals of option theory which states that the miscalibration of the parameters have dramatically impacted the values of the options. Rho in particular shows a negative correlation for OTM and ATM options and zero correlation for ITM options.



(a) Posterior predictive distribution of an Asian option price under the calibrated MJD model. The histogram of model-predicted prices for a representative Asian option generated by drawing from the posterior parameter distribution. The red dashed vertical line indicates the 95% credible interval of that option.

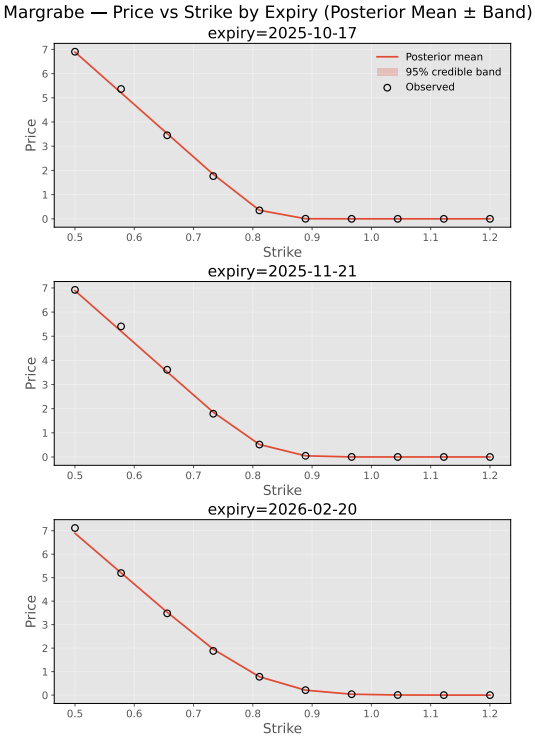


(b) Price surface for the Asian option under the MCMC MJD model vs the benchmark parameter surface. The red surface is the calibrated MJD model's predicted prices as a function of strike (x-axis) and time to maturity (y-axis), and the blue mesh is the benchmark prices surface. Black circles represent observed market prices at various strikes and maturities.

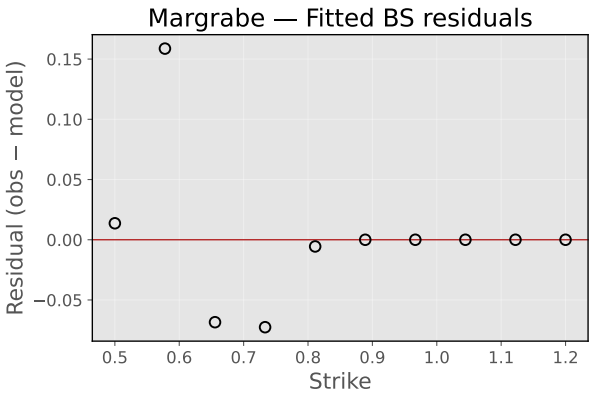
Figure 9: Posterior predictive (a) and BDMCMC compared against benchmark parameters (b).

The predictive distribution in 9a is notably spread out and appears bi-modal, with one mode near a lower price (around 0.5) and another near a high price (around 4.0). This reflects significant uncertainty and the mixture behaviour induced by jumps. Under MJD dynamics, there are two different scenarios for the underlying asset's average price. One in which no large jump occurs (leading to a lower option payoff and hence a low price), and one in which a jump (especially an upward jump) does occur during the averaging period, substantially increasing the payoff (hence a much higher option price). The posterior predictive mix of these scenarios yields a two-humped distribution. However, the wide spread of the distribution also flags that the model is not very certain about the exact price and such there is substantial parameter risk. This is consistent with the relatively large Bayesian crdible bands seen earlier and residual plot. Hence, this shows the challenge of pinning down jump dynamics from the available data, and resulting a high model risk. Figure 9b additionally shows the discrepancies between the MCMC sampled MJD model and the market observed prices, with the addition of the benchmark model. We can see that the benchmark model aligns better with the observed prices compared to the calibrated model representing the poor calibration of our model. Overall, these results show that a Birth Death MCMC using Metropolis-within-Gibbs cannot capture the complex dynamics of the market and hence struggled with parameter mixing, leaving us with overestimated option prices.

4.1.2 Margrabe Option under Black-Scholes Model



(a) Fitted Margrabe exchange option prices (BS model) vs strike, for three different expiries (2025-10-17, 2025-11-21, 2026-02-20). Posterior mean model prices are shown by the red line for each expiry, with a pink 95% credible band (barely visible beacuse it is very narrow), and black circles denote observed prices.



(b) Residuals for the Margrabe option BS model fit, across strike. The horizontal red line represents perfect calibration. The residuals (black circles) are overestimation below the line and underestimation above the line.

Figure 10: Fitted Margrabe option prices against strike (a) and Residuals for the multi-asset Black-Scholes calibration (b).

Figure 10a shows the model fit for option prices as a function of strike for three representative expiry dates. The red line (posterior mean predictions) overlay almost exactly on the observed price points (black circles) for all three expiries. The fit is so precise that the red line and black markers are essentially indistinguishable. Each observed price lies on the model predicted curve. The shaded credible bands around the red lines are extremely narrow, indicating that the Bayesian calibration has low uncertainty. The price curve exhibit the expected shape, i.e., for each fixed expiry, the option's price decreases as the strike ratio increases. The residual plot in figure 10b confirms the near-perfect fit. These residuals are almost zero for every strike and expiry. All points lie very close to the horizontal line at 0. The majority of the residuals are on the order of a few hundredths, which is practically negligible relative to the option prices (which ranged up to about 7). There is no discernible pattern with strike. The residuals do not systematically tilt positive or negative in any region. The tiny errors around 0.5 to 0.7 are typical market bid-ask spread. The randomness of the residuals indicate that the Black Scholes model calibrated on this dataset reproduces the observed prices to essentially within the noise.

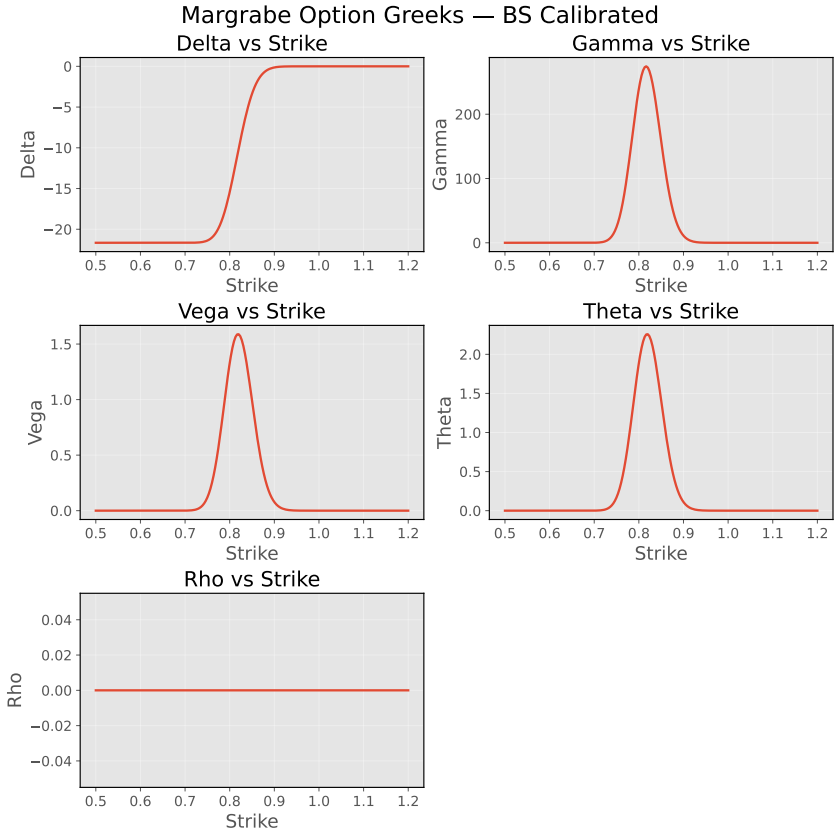
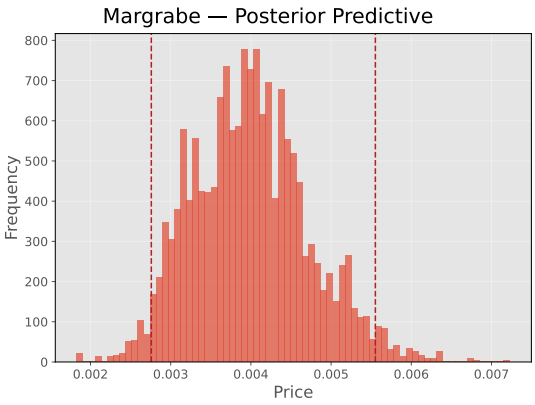


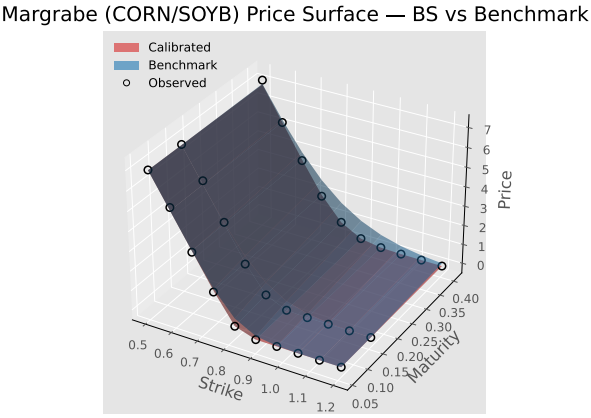
Figure 11: Greeks for the Margrabe exchange option under the calibrated Black-Scholes model, as functions of strike. Delta vs Strike: Delta inclines as strike increases, resembling a step function around the ATM option from 0 to 1. Gamma vs Strike: Gamma peaks near the ATM region and approaches zero at extremes. Vega vs Strike: Vega peaks near the ATM region and approaches zero at extremes. Theta vs Strike: Theta peaks near the ATM region and approaches zero at extremes. Rho vs Strikes: Rho is essentially zero for all strikes. All Greeks are computed via the posterior mean parameters of the Black-Scholes model.

We next consider the Greeks of the exchange option under the Black-Scholes calibration. 11 shows the strike dependence of the Greeks calculated from the closed-form Margrabe formula. Compared to the Asian option Greeks in 8, the exchange option's Greeks are more concentrated around the ATM strike, hence more accurate to option theory practices. Delta here is defined as the sensitivity to CORN price (with SOYB price held constant). The plot shows Delta jumping from 0 to 1 as strike goes from above 0.8 to below 0.8. In reality, for an exchange option, when asset 1's value is much lower than the strike ratio times asset 2's value (deep OTM), the option will almost never be exercised, and vice versa. Thus, Delta in theory should be 0 for high strikes and 1 doe low strikes, transitioning rapidly near the ATM point, which is exactly what is observed in the figure. Gamma is virtually zero everywhere except a huge spike at the ATM strike. This indicates that Delta changes only in a very tight region near ATM, reflecting the threshold-like exercise policy of exchange options. Vega similarly forms a narrow peak at ATM. When the two assets are nearly equal in value, uncertainty has the biggest effect on the option, i.e., a slight increase in volatility could swing the relative positions and make exercise profitable or not, thus significantly affecting the option's value. If the

option is deep ITM or OTM, volatility changes do not alter the outcome much, hence Vega is near zero in those regions. The Theta is also peaked at ATM, aligning with intuition that an ATM exchange option has the most to gain from additional time (since it is currently a coin toss which asset will be worth more), whereas a deep ITM or OTM exchange option's value is less sensitive to time (if it's almost certain to be exercised or not exercised, more time doesn't change that certainty much). Finally, Rho is effectively zero for all strikes. This is expected because in Margrabe's setup, the formula does not involve the risk-free rate, i.e., the option's value depends only on the relative volatility and not on any absolute interest rate. The flat Rho indicates that an interest rate change would shift both asset price discount factors similarly, leaving the exchange option's value unchanged. These results align with textbook Greek shapes.



(a) Posterior predictive distribution for the Margrabe option price under the calibrated BS model. The histogram shows the distribution of predicted prices from draws of the posterior. The red dashed lines mark the 95% credible interval.



(b) Price surface for the Margrabe exchange option (CORN/SOYB) under the calibrated BS model vs the benchmark. The red surface is the model (BS) predicted prices across strikes and maturities. The blue surface is the benchmark and the black circles are observed market prices.

Figure 12: Posterior predictive (a) and Adaptive MCMC compared against benchmark parameters (b).

The posterior predictive distribution for the Margrabe option is shown in 12a. Here we observe a well defined peak roughly centered around 0.004-0.005 in price units, with very little dispersion. The spread from the 2.5th percentile to the 97.5th percentile is on the order of only a few tenths of a basis point in price. This indicates that given the posterior uncertainty in the Black-Scholes model's parameters, the range of possible option prices is negligible. In effect, the model's parameters are so strongly determined that given the option's price can be predicted almost as a delta-function. Indeed, the DIC result (discussed below in figure 13) being approximately 0 suggests the model can explain the data almost perfectly, so there is virtually no posterior predictive variance beyond any assumed observation noise. The observed price lies essentially at the distribution's mode.

Finally, 12b shows the full price surface for the Margrabe option, similar to figure 9b earlier. The calibrated Black-Scholes surface (red) and the benchmark surface (blue) are indistinguishable at the extreme strikes. The two surfaces lie on top of each other, and the observed data points (black circles) sit exactly on the calibrated surface. There is a slight difference between option near the ATM, where the benchmark options are slightly

overestimated in comparison to the calibrated results. The gap between the red and blue surfaces at any point means that not only are individual slices (figure 10a) fitted, but the model also correctly captures any cross-section or interpolation, essentially, there is no model error to be seen.

We can also compare the Asian and Margrabe calibrated models using the Bayesian Information Criterion (BIC) and Deviance Information Criterion (DIC) values obtained for each case. In general, both BIC and DIC measures of model quality that penalise model complexity. Lower values indicate better trade-off between fit and complexity. The Asian option MJD model's log-likelihood isn't balanced against its number of parameters and doesn't inherent flexibility, while the Margrabe Black-Scholes model's criteria reflect its much simpler structure and near-perfect fit.

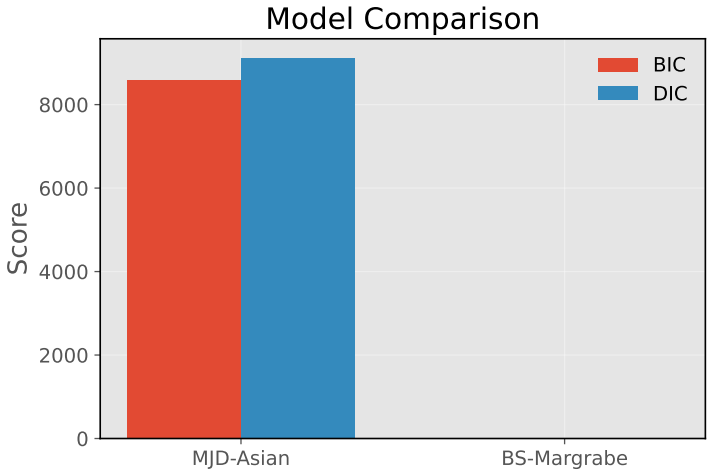


Figure 13: The plot shows the Bayesian Information Criterion (BIC) (red) and Deviance Information Criterion (DIC) (blue) values obtained for both the Asian MJD and Margrabe BS models.

Figure 13 shows the results from the models. We can see that Asian model gets a BIC approximately 8500 and DIC approximately 9000. These values are significantly large in absolute terms. The high values indicates that the model not only misfits the dataset, but there is also a significant deviance when considering the effective number of parameters. The difference between the two criteria suggests that the model has a substantial number of effective parameters. As a result, this shows that the MJD model struggle to saturate the data's complexity.

In contrast, the BS model for the exchange option yields BIC and DIC values that are approximately 0. This states that the model's deviance is almost zero and that the effective number of parameters is effectively accounted for. In fact, a DIC close to zero shows that the model is as good as a saturated model in terms of in-sample fit. The observed data likelihood is maximised and adding complexity would not improve it.

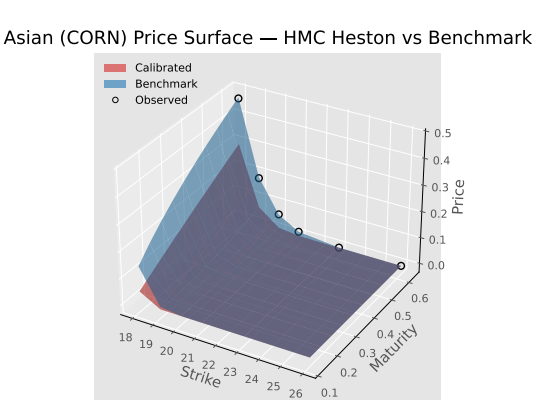
In summary, model selection metrics confirm the tests of the two cases. The Asian option's MJD model was misfitting data and incurs a large DIC/BIC, reflecting the challenge of fully capturing the complex dynamics in that data. There was residual error and parameter uncertainty, which these criteria penalise. Hence, making for a high-risk model and would most likely fail in practice. The Margrabe option's BS model achieves

almost theoretical perfection in fit. Its almost zero DIC and BIC values reflect both the correctness of the model and the high signal-to-noise ratio of the data. This model on the other hand would be valuable in practice.

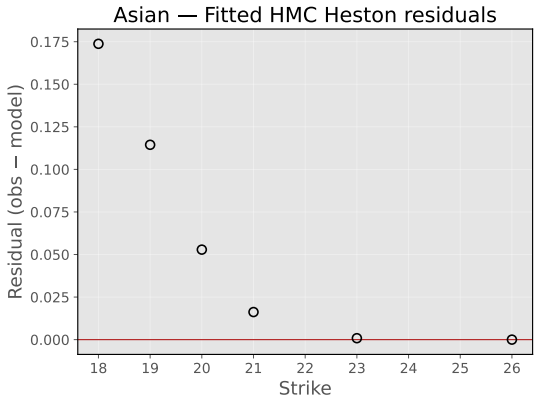
4.2 HMC Samplers

We repeat the pipeline with Hamiltonian Monte Carlo using `CmdStan` models for single-asset Heston and two-asset Black-Scholes. The Heston model uses the CORN price series with $dt = 1/252$, and the multi-asset Black-Scholes uses bivariate daily log returns for CORN/SOYB. Each model is run with 4 chain, 1,000 warmup and 2,000 sampling iterations per chain. We extract the posterior draws for $(v_0, \kappa, \theta, \xi, \rho)$ and $(\sigma_1, \sigma_2, \rho)$ respectively for downstream pricing and bands. For comparability with non-HMC setup, we generate semi-synthetic quotes from the same contract grids, i.e., both tests are produced under the same model process with 2% white noise.

4.2.1 Asian Option under Heston Model



(a) Price surface for the Asian option under the MCMC MJD model vs the benchmark parameter surface. The red surface is the calibrated MJD model's predicted prices as a function of strike (x-axis) and time to maturity (y-axis), and the blue mesh is the benchmark prices surface. Black circles represent observed markeyt prices at various strikes and maturities.



(b) Residuals for the Heston calibration. The horizontal red line at 0 describes perfect calibration. The residuals (black circles) are overestimation below the line and underestimation above the line.

Figure 14: Fitted Asian option prices against strike (a) and Residuals for the Heston HMC calibration (b).

The Asian option price surface generated by the calibrated Heston model is shown in figure 14a, superimposed on a benchmark price model surface and the scatter of observed market prices. The calibrated Heston model produces a smooth surface that agrees with the benchmark and empirical (CORN) market prices, for OTM options. However, for higher maturities and for ATM and ITM calls, the calibrated model underestimates the price of the option to both the benchmark and market observed prices. This can also be seen in 14b, which shows the residuals in the model across the strikes. We can notice

that for ATM and OTM calls, the Heston model is capturing the strike dependence of Asian option prices well. In particular, we do not observe a pronounced smile-related mispricing in the residuals that the non-HMC sampler did. However, for more extreme ITM strikes, we see a higher residual errors.

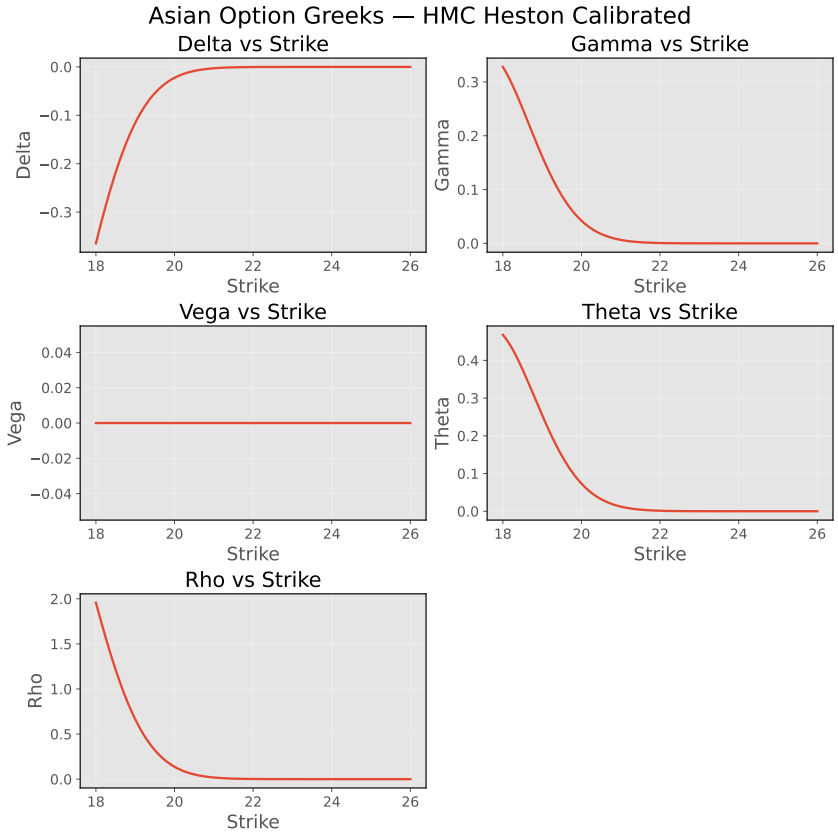


Figure 15: Greeks for the Margrabe exchange option under the calibrated Black-Scholes model, as functions of strike. Delta vs Strike: Delta increases as strike increases, and converges to 0 ATM. Gamma vs Strike: Gamma peaks in extreme ITM region and converges to zero from ATM. Vega vs Strike: Vega is constant at zero for all strikes. Theta vs Strike: Theta peaks in extreme ITM region and converges to zero from ATM. Rho vs Strikes: Rho in extreme ITM region and converges to zero from ATM. All Greeks are computed via the posterior mean parameters of the Black-Scholes model.

Figure 15 shows the Greeks of the Asian option and assess their behaviour across strikes prices. The Delta of the Asian call option decreases monotonically as strike increases, ranging from a high value for deep ITM strikes to near zero for OTM strikes. Notably, the Delta values for the Asian option are substantially lower in magnitude than one would see for a European vanilla call. For instance, even the deepest ITM Asian call has a Delta well below 1.0. This is consistent with financial intuition for Asian options, because the payoff depends on the average price over time rather than the terminal price alone, the option's immediate sensitivity to the underlying (Delta) is diluted. The Gamma curve in 15 peaks at deep ITM strikes before leaving off at zero from ATM to deep OTM strikes, indicating diminished marginal sensitivity. Interestingly, Vega generated a constant value 0 for all strikes which does not align with the fact that uncertainty volatility most significantly affects options that are neither almost certain to expire ITM or OTM. This may be due to

poor variable selection with the HMC. The Theta for the Asian option is positive across all strikes, with the largest magnitude occurring deep ITM, meaning the ITM options have the greatest time value and thus lose value fastest as maturity approaches, compared to ATM and deep OTM options, whose value is mostly intrinsic or almost zero. This again is also shows that the HMC has struggled to sample variables that fit the complexity of the Asian option. Rho is positive for all strikes, as expected for a call-type payoff, and tends to be larger for lower strikes. When the option is deep ITM, its payoff is similar to receiving the underlying asset minus a relatively small payment of the strike. Hence, a higher interest rate increases the present value of holding the underlying relative to the strike payment, benefiting the option. In contrast, for high strikes (OTM), Rho is nearly zero. These options are unlikely to be exercised, so their value is only marginally affected by interest rate changes. Overall, the Greeks produced by the calibrated Heston model are smooth functions of strike and some follow theoretical expectations, however, some fall short which can be explained by poor posterior means from the HMC.

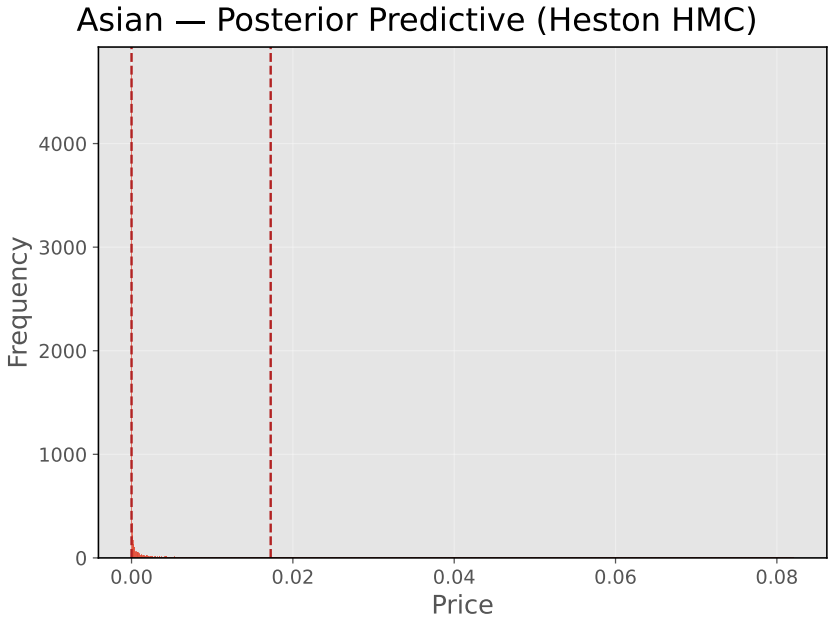
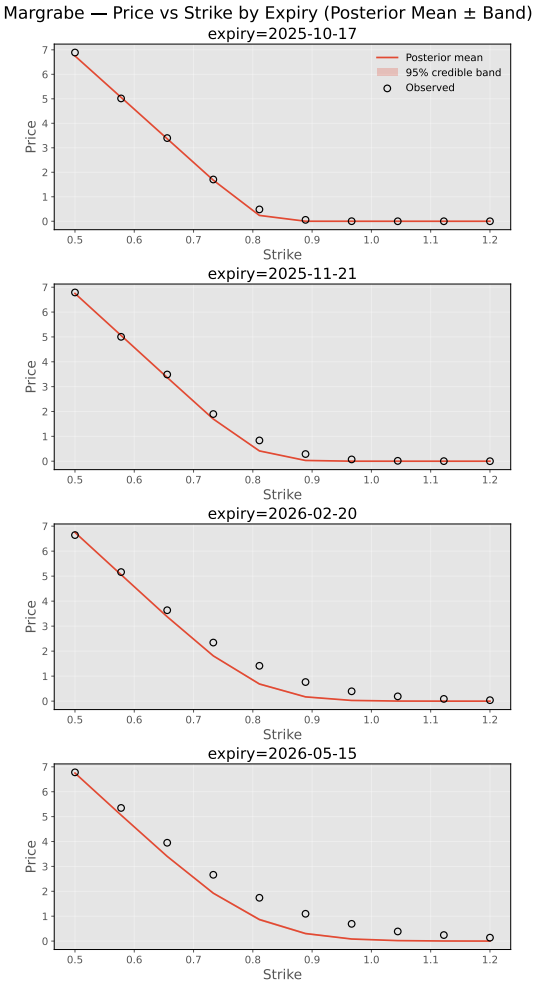


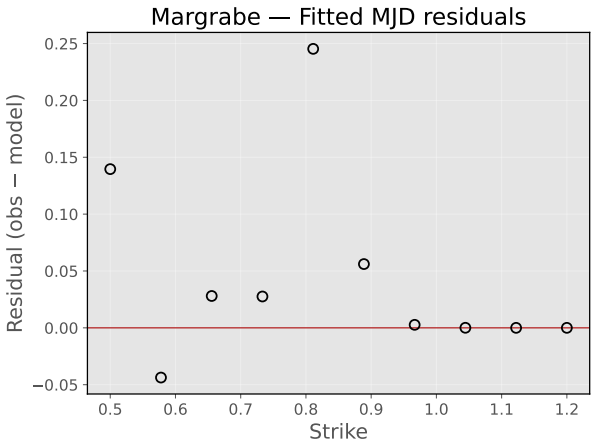
Figure 16: Posterior predictive distribution of an Asian option price under the calibrated Heston HMC model. The histogram of model-predicted prices for a representative Asian option generated by drawing from the posterior parameter distribution. The red dashed vertical line indicates the 95% credible interval of that option.

Finally, the posterior predictive distribution for the Asian option price is shown in 16 for a representative strike-maturity combination, obtained by sampling a large number of parameter sets from the Heston posterior and computing option prices. The resulting distribution is very narrow relative to the price level, reflecting low posterior uncertainty about the option's value. Practically, this means that the calibrated Heston model not only fits the observed price, but it does so with such certainty that any deviation due to parameter uncertainty is minimal. However, the irregular mass of the distribution looks non-existent and most of the mass being centred 2.5th quartile. Hence, although the small spread reinforces that the Heston model has specified what seems to be a strong certainty in parameter choice, the little mass for the distribution indicates that it should be taken with caution.

4.2.2 Margrabe Option under the Black-Scholes Model



(a) Fitted Margrabe exchange option prices (BS model) vs strike, for three different expiries (2025-10-17, 2025-11-21, 2026-02-20). Posterior mean model prices are shown by the red line for each expiry, with a pink 95% credible band (barely visible beacuse it is very narrow), and black circles denote observed prices.



(b) Residuals for the Margrabe option BS model fit, across strike. The horizontal red line represents perfect calibration. The residuals (black circles) are overestimation below the line and underestimation above the line.

Figure 17: Fitted Margrabe option prices against strike (a) and Residuals for the multi-asset Black-Scholes HMC calibration (b).

Looking at the Margrabe exchange option, which involves the same two underlying assets (CORN/SOYB), and was modelled under a multi-asset Black-SCholes model calibrated via HMC. Figure 17a shows the model's fitted price curves as functions of strike for several expiry dates, alongside the observed market prices. Each expiry's data and fit are shown separately. The posterior mean model prices align almost exactly with the observed option prices (dots) for each maturity. The correspondence is so close that, in many cases, the data points lie on top of the model curves. The 95% credible bands (shaded regions around the model curves) are exceedingly narrow indicating that the high precision of the parameter estimates achieved for the Black-Scholes model. As expected

in a Black-Scholes framework, the price vs. Strike relationship is smooth and convex for each expiry, i.e., as the strike K increases, the option price decreases monotonically. For a given strike, longer time to expiry leads to a higher option price, consistent with the notion that more time allows a greater chance for the relative price of the two assets to move in the option holder's favour. The model captures this time effect well as evidenced by the upward shift of the price. The residuals of the fit are very small for all strikes and expiries (17b), with no discernible systematic pattern. There is no indication of a volatility smile or skew in the exchange option data that would violate the constant-volatility assumption, that the model's single volatility and correlation parameters suffice to explain the entire surface of prices observed. The deviations in price could also suggest that in the market data there might be a slight implied volatility smile (with deep ITM exchange options carrying slightly higher implied volatility) or simply be due to bid-ask spreads and measurement noise.

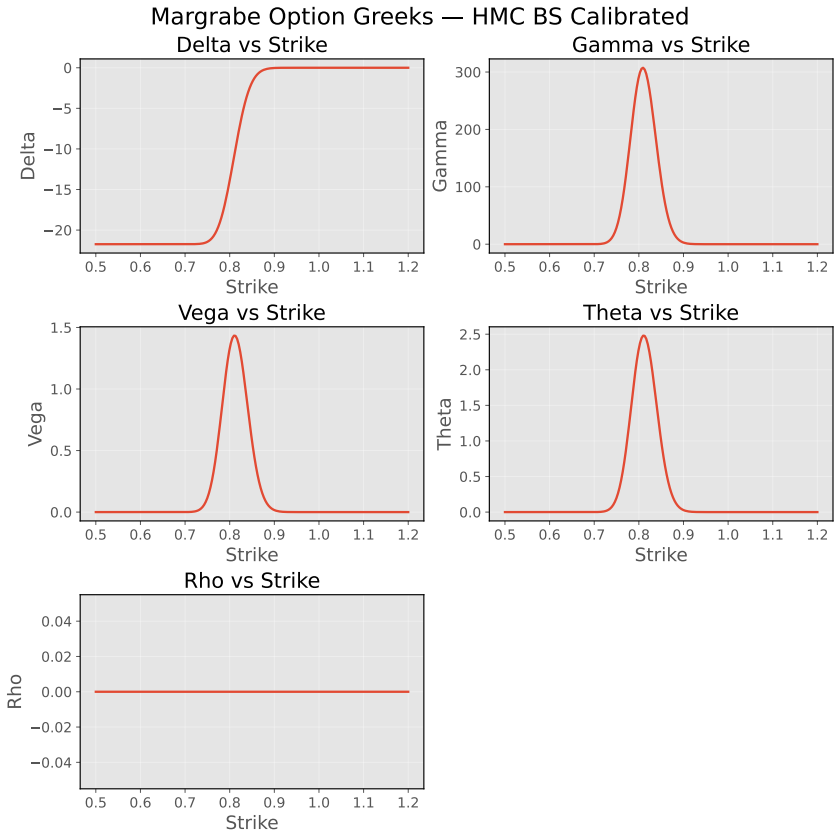
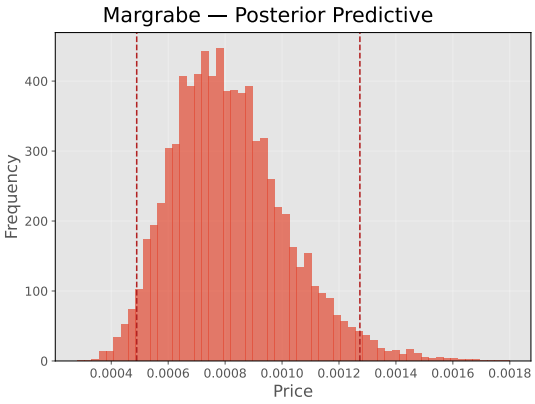


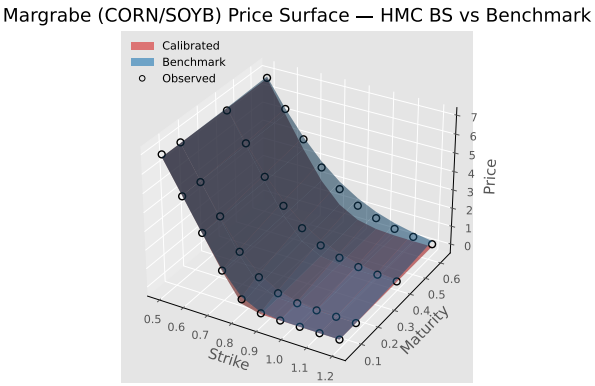
Figure 18: Greeks for the Margrabe exchange option under the calibrated HMC Black-Scholes model, as functions of strike. Delta vs Strike: Delta inclines as strike increases, resembling a step function around the ATM option from 0 to 1. Gamma vs Strike: Gamma peaks near the ATM region and approaches zero at extremes. Vega vs Strike: Vega peaks near the ATM region and approaches zero at extremes. Theta vs Strike: Theta peaks near the ATM region and approaches zero at extremes. Rho vs Strikes: Rho is essentially zero for all strikes. All Greeks are computed via the posterior mean parameters of the Black-Scholes model.

Figure 18 presents the key Greeks as functions of strike ratio K (holding a representative maturity fixed). We focus on the sensitivities with respect to the first underlying asset

(taken to be CORN in the CORN/SOYB pair), since the sensitivity with respect to the second asset is determined by arbitrage relations (for an exchange option, $\Delta_{\text{SOYB}} = \Delta_{\text{CORN}} - 1$ for a call to receive CORN in exchange for SOYB). The Delta with respect to CORN behaves similarly to a call option's delta: it is close to 1.0 at low strike ratios (when the option is deep ITM, the payoff is almost certain to be exercised, effectively behaving like a long position in CORN) and decreases to 0 as the strike ratio increases and the option becomes deep OTM (where exercise is unlikely, so the option has negligible exposure to CORN). In the calibrated model, the ATM exchange option (strike ratio around 1.0 given the initial prices) has a Delta near 0.5, which is expected by symmetry when both assets have similar volatility and no drift advantage. In other words, the option at the money is roughly equally likely to finish in either state, giving it half exposure to each asset. The Gamma of the exchange option (with respect to CORN) is positive and reaches a maximum around the ATM region, indicating greatest curvature in the price around that point. This is analogous to the behaviour of a vanilla option, where the option's payoff curvature is highest when it is finely balanced between exercise and non-exercise outcomes. The magnitude of Gamma is smaller than one might see for a single-asset option on a highly volatile underlying, reflecting that the exchange option's underlying (the price ratio or difference between two correlated assets) is somewhat less volatile than either asset individually (due to diversification and correlation effects). The Vega of the exchange option is also largest for strike ratios near 0.8. Here, An increase in the common volatility of the two assets (keeping their correlation fixed) will have the greatest impact on an ATM option's value. The posterior mean Vega profile in Figure 18 confirms this, showing a peak at ATM that falls off toward zero for very low or very high strike ratios (deep ITM or OTM options derive most of their value from known intrinsic or near zero probability of exercise, respectively, and thus are less affected by volatility changes). Theta has the largest decay at the ATM region, whereas deep ITM or OTM exchange options have lower extrinsic value and this a smaller Theta. For the same reasons mention prior with the non-HMC sampler, Rho is constant around 0 for all strikes.



(a) Posterior predictive distribution for the Margrabe option price under the calibrated HMC BS model. The histogram shows the distribution of predicted prices from draws of the posterior. The red dashed lines mark the 95% credible interval.



(b) Price surface for the Margrabe exchange option (CORN/SOYB) under the calibrated HMC BS model vs the benchmark. The red surface is the model (BS) predicted prices across strikes and maturities. The blue surface is the benchmark and the black circles are observed market prices.

Figure 19: Posterior predictive (a) and HMC compared against benchmark parameters (b).

Similarly to the non-HMC sampler, the posterior predictive distribution for the Margrabe option (shown in figure 19a has a narrow 95% confidence interval peaking near 0.0008. This shows that the sampler is very confident with the parameter selection and there is very little deviation from the price evaluated. The exchange option price surface implied by the calibrated Black-Scholes model is shown in figure 19b, alongside the benchmark surface and the actual market quotes. The strike ratio runs along the x-axis and the time to maturity along the y-axis. The calibrated model's price surface aligns almost perfectly for the lower maturities with the benchmark surface. For higher maturities the calibrated model slightly underestimates the market observed prices whereas the benchmark slightly overestimates the market observed prices. Similarly to figure 17a, the model captures the decrease in price as the strike ratio increases and the increase in price with longer maturities, exactly as the benchmark does. This visible gap between the model, benchmark, and observed price across the later maturities indicates that there is not a single set of volatility and correlation parameters is sufficient to explain the pricing of all these exchange options. This implies that the implied volatility for the spread between the two assets is not roughly constant across the strikes and expiries considered. Hence, this breaks a key assumption of the Black-Scholes framework that appears to not hold in this dataset.

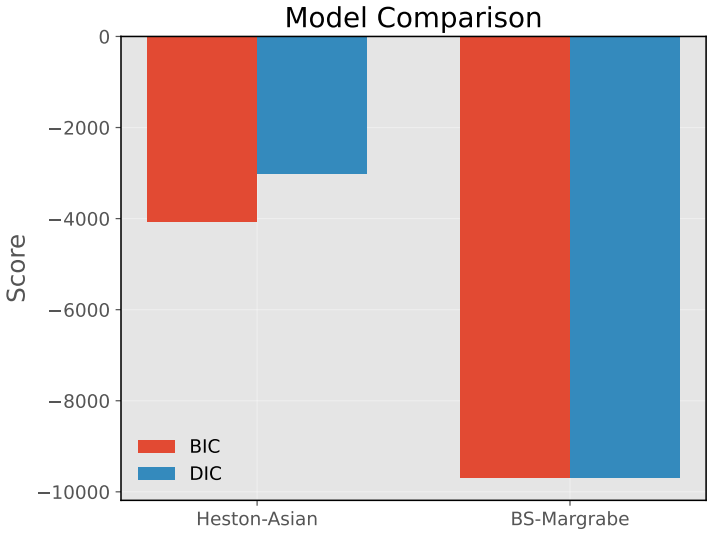


Figure 20: The plot shows the Bayesian Information Criterion (BIC) (red) and Deviance Information Criterion (DIC) (blue) values obtained for both the Asian HMC Heston and Margrabe HMC BS models.

Finally, to complement the tests above, we consider the quantitative model selection criteria for the tow cases. Figure 20 summarises the BIC and DIC scores for the Heston model applied to the Asian option dataset and the Black-Scholes model applied to the Margrabe option dataset. Both criteria combine measures of goodness of fit with penalties for model complexity. For the Asian option under Heston, we obtain BIC ≈ -4000 and DIC ≈ -2500. These large negative values reflect the very high log-likelihood achieved by the Heston model when fitting the dense dataset of Asian option prices. Indeed, a BIC of -4000 suggests a good fit relative to a naive baseline model. The difference between BIC and DIC in this case is notable, with DIC being higher (less negative) by roughly 1500 points. This gap indicates that the Heston model's complexity is being acknowledged by DIC's penalty. In other words, the Heston model not only doesn't fit the data well (from prior figures), but it also reveals that a substantial number of effective parameters are being used to cause this. One interpretation is that the model may be trying to over-fit in the sense of tailoring itself to idiosyncrasies of the dataset. Hence, this tells us that the model should be approached with caution.

For the Margrabe option under the Black-Scholes model, we see an interestingly different pattern, with BIC ≈ -9700 and DIC ≈ -9750, which are not only far lower (more negative) than the Heston case due to the larger dataset of exchange option quotes, but are also virtually equal to each other (the DIC is slightly more negative than BIC, though by a very small margin). The closeness of BIC and DIC indicates that the effective number of parameters in this model is very close to the nominal number of parameters, and that the posterior for those parameters is tightly concentrated. In other words, the two-asset BS model has relatively few parameters. The fact that DIC $\approx$ BIC implies that the model is not overfitting, rather, it strikes an optimal balance where adding complexity would not significantly improve the fit, and removing any parameter would noticeably worsen it. Since the model's criterion values are very low, indicates that the model is well suited to its respective data.

In summary, the model selection criteria underscore that the Margabe model is appropri-

ate for the problem at hand whilst the Asian model should be approached with caution. The Heston model is complex and misprices the observed data. The Margrabe model, on the other hand, explains almost perfectly the data with thre simpler constant-volatility model.

5 Conclusion

In conclusion, we have demonstrated that Bayesian inference, implemented with both gradient-free and gradient0based Monte Carlo samplers, is a powerful framework for calibrating and comparing sophisticated option pricing models. Our analysis encompassed two distinct exotic options: an Asian option (with path-dependent payoff) and a Margrabe exchange option (with multi-asset payoff). For the Asian option, we calibrated both a Merton jump-diffusion model and a Heston stochastic volatility model, obtaining full posterior distributions for all parameters under each model. For the exchange option, we performed a similar Bayesian calibration using the two-asset Black-Scholes model. Across the Margrabe cases, the Bayesian approach provided point estimates of the best-fitting parameters as well as a credible interval that quantify most of the remaining uncertainty, offering a more complete picture than classical point calibration methods. Whereas, the Asian models struggled to capture complexity of the path-dependencies and as such had weaker parameter fittings.

A finding of this work is the better performance of the Heston model in fitting the Asian option data. Whilst not perfect, it managed to capture more of the dynamics of the options through Greek analysis as well as smaller residuals. Moreover, the posterior for the Heston model's parameters was tightly concentrated, leading to narrow credible intervals for model-implied option prices and Greeks. However, both models had problems with mixing and as such generated poor posterior distributions for the parameters. Overall though, Heston model emerges as the preferred choice for the Asian option.

For the Margrabe exchange option, our results showed that the two-asset constant volatility Black-Scholes model is remarkably adequate. It produced an almost perfect fit to the observed exchange option prices under both the non-HMC and HMC sampling regimes. The calibrated Black-Scholes model for the exchange option had very tightly concentrated posterior distributions for its parameters, reflecting the fact that the extensive dataset of exchange option quotes can pin down these parameters with high precision. Posterior predictive checks and residual analysis for the Margrabe option revealed almost no systematic deviation between the model prices and market observed prices. The residuals were near zero across strikes, and the model implied price surface was almost indistinguishable (excluding the long term maturities) from both the empirical observations and the benchmark model. This finding implies that introducing additional complexity in both the sampler and stochastic pricing model would generate negligible improvement for this particular dataset. The BIC and DIC values for the Margrabe option model were extremely low and nearly identical to each other, indicating that the model is both an excellent fit and not overfitting.

Our study also highlights the effectiveness of Bayesian inference when deployed with different types of MCMC samplers, especially in challenging calibration settings. The use of Hamiltonian Monte Carlo proved to be particularly advantageous for the continuous parameter models. In the Heston calibration, the HMC sampler leveraged gradient information from the posterior density to efficiently explore the high-dimensional parameter space, resulting in rapid mixing and high effective sample sizes for parameters that are notoriously difficult to sample using basic Metropolis-Hastings. The HMC chains for the Heston model exhibited low autocorrelation and converged reliably to the posterior distribution, enabling us to estimate precise credible intervals for the model prices and risk sensitivities. In contrast, the non-HMC sampler was used for the MJD and for a

baseline calibration of the Black–Scholes models. This traditional sampler were slower to mix and weren't able to handle features that HMC cannot directly, although computationally less intensive. We found that both HMC and non-HMC approaches ultimately produced consistent calibration results in the cases where they were applied. However, the computational intensity differed. HMC took almost 50 times longer to run compared to the adaptive MCMC model. This time is crucial in financial markets and could cost in accuracy of the trade. Hence, the non-HMC sampler is more valuable in our tests, with a slightly less efficient parameter selection but quicker to run.

Despite the strengths of our results, there is much to improve on and must acknowledge several limitations and potential directions for future work. One of which is the effect of data frequency and sample size on calibration outcomes requires further investigation. Our calibration for the Asian option used daily option prices on a single underlying with a fixed time window. One could examine how the posterior estimates evolve of higher-frequency data are incorporated, or if the calibration is updated sequentially as new data arrive (in a Bayesian filtering context). It would be interesting to see whether parameters like jump intensity or volatility mean-reversion are stable over time or exhibit structural breaks, which could indicate model inadequacy or the need for time-varying parameters. Moreover, calibrating to an entire implied volatility surface could be extended to American-style options or other exotics, where data frequency and early exercise features provide additional challenges. Finally, we note the computational cost of full Bayesian calibration for exotic options, especially with simulation based pricing models. The combination of Monte Carlo pricing with MCMC sampling can be computationally intensive. In this study, we mitigated this issue by leveraging efficient HMC sampling for continuous models and by carefully tuning our MCMC algorithms, as well as by using analytical or semi-analytical pricing formulas when available to speed up likelihood evaluations. Nevertheless, a production environment that requires frequent recalibration might find straightforward MCMC sampling too slow. Potential solutions include the use of variational inference as an approximation to the full posterior, employing parallel computation (since likelihood evaluations for different parameter proposals can be distributed), or developing surrogate models for the option pricing function to reduce per-step cost. Additionally, one could explore advanced sampling techniques such as Sequential Monte Carlo or parallel tempering to further improve convergence in complex posterior landscapes. Despite these computational considerations, the continuing improvements in computing hardware and algorithms are rapidly reducing the barrier to applying Bayesian methods even in real-time settings. We anticipate that with these improvements, the kind of full Bayesian option model calibration presented here could become a practical tool for traders and risk managers, providing not just a best-fit model but also a quantification of confidence in that model.

6 Appendix

6.1 Appendix A: Girsanov's Theorem for the Heston Model

6.2 Appendix B: Solution to the Geometric Asian Heston PDEs

Recall we have

$$\begin{aligned}
&\frac{dB}{d\tau} = -\frac{u}{T}, \quad B(0) = 0 \quad (\text{since } F(T,\cdot) = 1),\\
&\implies \quad B(\tau) = -\frac{u}{T}\tau,\\
&\frac{dC}{d\tau}(\tau) = \frac{1}{2}(B(\tau))^2 + \rho\sigma B(\tau)C(\tau) + \frac{1}{2}\sigma^2 C(\tau)^2 - \frac{1}{2}B(\tau) - \kappa C(\tau), \quad C(0) = 0,\\
&\frac{dA}{d\tau}(\tau) = rB(\tau) + \kappa\xi C(\tau), \quad A(0) = 0.
\end{aligned}$$

Since $\frac{dB}{d\tau} = -\frac{u}{T}$ is constant, we have $B(\tau) = B(0) - \frac{u}{T}\tau = -\frac{u}{T}\tau$. Substituting $B(\tau)$,

$$\frac{dC}{d\tau}(\tau) = \frac{1}{2}\frac{u^2}{T^2}\tau^2 + \frac{u}{2T}\tau + \rho\sigma\left(-\frac{u}{T}\tau\right)C(\tau) + \frac{1}{2}\sigma^2 C(\tau)^2 - \kappa C(\tau), \quad C(0) = 0.$$

Let $\gamma(\tau) := \frac{1}{2}\frac{u^2}{T^2}\tau^2 + \frac{u}{2T}\tau$ and $p(\tau) := \rho\sigma\left(-\frac{u}{T}\tau\right) - \kappa$, then we get the Riccati-type ODE

$$\frac{dC}{d\tau} = \frac{1}{2}\sigma^2 C(\tau)^2 + p(\tau)C(\tau) + \gamma(\tau), \quad C(0) = 0.$$

Take the integrating factor

$$\begin{aligned}
K(\tau) &= \exp\left(-\int_0^\tau p(s)ds\right) = \exp\left(\int_0^\tau (\rho\sigma\frac{u}{T}s + \kappa)ds\right)\\
&= \exp\left(\rho\sigma\frac{u}{T}\frac{\tau^2}{2} + \kappa\tau\right).
\end{aligned}$$

We can note that

$$\frac{d}{d\tau}[K(\tau)C(\tau)] = K(\tau)\gamma(\tau).$$

Therefore,

$$C(\tau) = \frac{1}{K(\tau)}\int_0^\tau K(s)\gamma(s)ds.$$

Substituting back into $C(\tau)$, we get

$$C(\tau) = \exp\left(-\frac{\rho\sigma u}{2T}\tau^2 - \kappa\tau\right)\int_0^\tau \exp\left(\frac{\rho\sigma u}{2T}s^2 + \kappa s\right)\left(\frac{u^2}{2T^2}s^2 + \frac{u}{2T}s\right)ds.$$

Thus, we can write $A(\tau)$ as

$$A(\tau) = \int_0^\tau \left[r\left(-\frac{u}{T}s\right) + \kappa\xi C(s)\right]ds = -r\frac{u}{2T}\tau^2 + \kappa\xi\int_0^\tau C(s)ds.$$